

RESEARCH ARTICLE

Advancing family dementia caregiver interventions in low- and middle-income countries: A pilot cluster randomized controlled trial of Resources for Advancing Alzheimer's Caregiver Health in Vietnam (REACH VN)

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Abstract

Introduction: Low- and middle-income countries have rapidly increasing numbers of people with dementia, yet little evidence on family caregiving interventions. We tested the preliminary efficacy and feasibility of a family caregiving intervention in northern Vietnam.

Methods: Nine clusters comprising 60 family caregivers were randomized to a culturally adapted version of a four- to six-session, multicomponent intervention delivered in-home over 2 to 3 months, or enhanced control. Eligible caregivers were ≥ 18 years of age and scored ≥ 6 on the Zarit Burden Inventory (ZBI).

Results: Fifty-one caregivers (85%) completed the study. Using analysis of covariance with 3-month assessment as the outcome and baseline assessment as a covariate, intervention group caregivers had an average ZBI (primary outcome) score 1.2

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standard deviation (SD) lower ($P = .02$) and Patient Health Questionnaire-4 (psychological distress) score 0.7 SD lower ($P = .03$) than controls.

Discussion: In the first study of its kind in Vietnam, a culturally adapted, manualized, family caregiver intervention was both efficacious and feasible.

KEYWORDS

Alzheimer's disease, dementia, family caregiving, global health, low- and middle-income countries, non-pharmacological interventions, Vietnam

1 | INTRODUCTION

Although Vietnam and other low- and middle-income countries (LMICs) face an enormous public health challenge as their populations age and the numbers of older adults with Alzheimer's disease and Alzheimer's-related dementias (AD/ADRDs) increase,^{1,2} many are not well-prepared to provide effective care and support to family caregivers. Because AD/ADRDs are among the most disabling non-communicable diseases, family members are often called upon to provide the bulk of day-to-day care, particularly in LMICs where nursing homes and other types of formal support services are scarce.³ Little is known, however, about which caregiver interventions are most effective and sustainable in LMICs, creating a critical evidence gap and barrier for policymakers.⁴

Supporting family caregivers through community-based programs is widely viewed as an essential part of national plans to address AD/ADRDs.⁵ This is also true for Vietnam, a country where very substantial gaps in diagnosis, care, and services for persons with dementia and their families exist in the healthcare system and efforts are underway to develop a national plan to help address these issues.⁶ The reason is that these programs and services are essential to mitigate the adverse social, economic, and health/mental impacts of caregiving that are well-documented in the United States and other high-income countries (HICs).⁷ These adverse impacts disproportionately affect women who provide the bulk of family AD/ADRD caregiving. In LMICs, families may be particularly vulnerable to the negative economic impacts of caregiving because many lack economic resources and access to long-term services and supports.

We are learning much more about how best to support family dementia caregivers in HICs. There is now a broad consensus, for example, that the most effective caregiver interventions are multi-component and often include education, skill-building to manage difficult behavioral problems, stress reduction, and referral to community resources.^{7,8} There is a need to understand whether these interventions are also effective in LMICs, where the resources and supports for caregivers may be more limited.

This study presents the results of the first clinical trial to address this critical evidence gap in Vietnam, a country with one of the world's fastest growing elderly populations.⁹ We adapted and tested a version of REACH (Resources for Enhancing Alzheimer's Caregivers Health),^{10,11} one of the most widely tested models in the United States,

which has been found to be effective and feasible in multicultural populations in the United States¹²⁻¹⁵ and internationally including Hong Kong.¹⁶ The specific aims of this study were to examine the preliminary efficacy and feasibility of a culturally adapted version of REACH (ie, REACH Vietnam) for family members of persons living with dementia in Vietnam.

2 | METHODS

2.1 | Study design and participants

From June to October of 2018, we conducted a **cluster randomized controlled trial** (RCT) among family caregivers of persons with AD/ADRDs who were identified through a registry of individuals who had been diagnosed previously with dementia by neurologists from the Vietnam National Geriatric Hospital (NGH) in collaboration with local commune health stations. Details of the study can be found in the published protocol.¹⁷ The study was registered with clinical trials.gov (identifier NCT03587974). Informed verbal consent was obtained from all participants and the protocol was approved by the institutional review boards of University of California, Davis, University of South Carolina, and the Vietnam NGH. The study was designed primarily as a feasibility study that also provided an opportunity to estimate effect sizes and to examine preliminary efficacy.

Participants resided in designated clusters (ie, geographic area served by a local health station) in Soc Son, a semi-rural district in Hanoi with a longstanding history of collaboration with the NGH. Soc Son is more economically advantaged compared with many other semi-rural areas due to its proximity to Hanoi and location near the international airport, an area undergoing considerable economic development. A convenience sample of 12 clusters was chosen based on their proximity to NGH, their participation in the NGH dementia registry, and history of successfully participating in prior research projects and initiatives. After the start of the study, we added as a criterion a minimum of three persons recruited from each cluster. This latter criterion was added based on resource considerations and the difficulty of estimating important study parameters (eg, inter-correlation coefficients) with too few participants. A cluster design was chosen because of concerns about possible contamination effects if participants in the control and intervention condition resided in the same community.

NGH staff contacted families of elders with dementia, introduced the study, and screened the caregivers for eligibility criteria. Caregiver inclusion criteria included age 18 or older, family member most involved in the dementia patient's day-to-day care, a score on the Zarit Burden Interview (ZBI) (4-item) of ≥ 6 ,^{18,19} and caring for an older adult with a dementia diagnosis and a Clinical Dementia Rating (CDR) score of 1 or above.²⁰ We excluded caregivers identified as having difficulties in the consent process due to cognitive impairment or severe sensory impairment. Caregivers meeting inclusion criteria were asked for permission to have research staff contact them to set up an enrollment visit. During the enrollment visit, which took place several weeks after the initial screening, research staff provided additional information for the study, completed verbal consent, and collected baseline data. Participants were reimbursed the equivalent of US \$9 for their participation in each session.

As part of the consent process, all caregivers were informed that this was a voluntary study and that they could withdraw at any time.

2.2 | Randomization and masking

Randomization was at the level of clusters, defined as geographic areas (ie, communes) served by commune health stations which have a population range of ≈ 5000 –15,000 people. Randomization was conducted by one of the study investigators residing in the United States through the flip of a coin.

At the time of consent, participants were informed about their allocation to either the enhanced control or the intervention condition. Research assistants who conducted the 3-month outcome assessment were masked to allocation. Baseline assessments were conducted by research staff who were not masked to allocation.

2.3 | Procedures

Participants in the intervention group received REACH VN, a culturally adapted version of the REACH VA (Department of Veteran Affairs) intervention.¹² REACH VA is based on REACH II²¹ and consists of four “core” training sessions on problem solving, mood management/cognitive restructuring, stress management (eg, signal breath, pleasant event scheduling), and communication, plus up to two additional sessions based on caregiver's needs and clinical judgment.^{11,12,21} In REACH VA, the sessions last 1 hour and are delivered by a trained interventionist by telephone, telehealth, or face-to-face.

Interventionists were trained and certified by a senior member of the research team (HN) who is bilingual and was herself certified by the REACH VA intervention by the Memphis Caregiver Center affiliated with the University of Tennessee. Training of interventionists in Vietnam consisted of didactic sessions based on the REACH VA training materials as well as “hands-on” field experience in a case series prior to the study onset in which interventionists worked directly with HN.

Our culturally adapted and manualized intervention, REACH VN, is based on the REACH VA intervention, with modifications to make

RESEARCH IN CONTEXT

- Systematic review:** As background to this study the authors conducted a scoping review of evidence for non-pharmacological interventions for family caregivers that had been conducted in low-, middle-, and high-income countries in Asia.⁴ Of 30 clinical trials identified by this search, only 4 were conducted in low- and middle-income countries and none in Vietnam (a low middle-income country). Although studies in high-income countries reported generally positive outcomes for caregivers, there was a striking gap in terms of evidence for low- and middle-income countries in Asia, which are projected to experience a rapid increase in the number of people living with dementia in coming decades.
- Interpretation:** Our findings show that a culturally adapted and relatively brief family caregiver intervention was both efficacious and feasible in Vietnam. Caregivers who received the intervention showed significantly reduced caregiver burden (primary outcome) and reduced psychological distress (secondary outcome) but no significant improvement in knowledge about Alzheimer's disease (secondary outcome). Effect sizes were relatively large for both caregiver burden and psychological distress. Our findings are novel, representing the first test of a family caregiving intervention in Vietnam and one of the few among low- and middle-income countries in Asia.
- Future directions:** Our results indicate the need for replication in a larger study under conditions that more closely resemble those in routine service delivery settings in Vietnam (eg, delivery of the intervention by local care providers and community outreach workers) to enhance sustainability at the local level and to guide policymakers as they seek models of support for caregivers that are promising for broader implementation and sustainability. Additional research is also needed to better understand the mechanisms that may explain the efficacy of this intervention and magnitude of the effect sizes among Vietnamese caregivers.

it suitable for the context of a semi-rural area in Vietnam. We used a multi-step cultural adaptation process guided by existing frameworks commonly used in global health.²² Although a full description of the adaptation process, modifications, and rationale is beyond the scope of this paper, the following is a summary of significant changes made to intervention content, context/delivery, and training.²³ Changes to content included numerous modifications of the intervention manual and caregiver notebook to make the scripts, examples, and resources appropriate to the culture and literacy level of the target population

(eg, substituting culturally relevant examples, simplifying language), and expanding the amount of time in across intervention sessions devoted to caregiver education about AD. Changes to context/delivery included participation of multiple family members in the intervention when appropriate, engaging the male head of the household in the initial session to facilitate participation and retention, and in most cases conducting weekly rather than biweekly sessions to sustain momentum. Changes to training included supplementing standard REACH VA training with principles of Buddhism to enhance interventionist skills and conducting a small case-series to give interventionists hands-on experience.²⁴

For this study, all interventionists were female healthcare and allied professionals including nurses, physicians, and social workers. Home visits were conducted by pairs of interventionists, one playing a lead role and the second a supportive role. Participants assigned to the enhanced control condition received a single face-to-face session at the time of enrollment in the caregiver's home (or another place of their choosing) that focused on education about the nature of dementia and included the provision of written educational materials. In addition, any safety issues identified during the single home visit were addressed by research staff.

2.4 | Study outcomes and their assessment

The objectives of this study were to assess both preliminary efficacy and feasibility of the intervention. For preliminary efficacy, the primary outcome was caregiver burden measured by the 4-item Zarit Burden Inventory (ZBI).¹⁸ Secondary outcomes were depression/anxiety symptoms measured by the Patient Health Questionnaire 4 (PHQ-4)²⁵ and Alzheimer's knowledge measured by the 30-item Alzheimer's Disease Knowledge Scale.²⁶ Primary and secondary outcomes were assessed at baseline and after completion of the intervention (≈ 3 months after enrollment). The feasibility outcomes were assessed in terms of cluster recruitment (% of clusters randomized that recruited minimum number of caregivers), individual recruitment (% of eligible caregivers who enrolled), retention (% of caregivers enrolled who completed study), intervention fidelity (% of sessions with delivery of all required activities based on standardized forms completed by interventionist), and assessments (% of participants with complete data). Baseline assessments were conducted at the time of enrollment by bachelor's level or higher research staff. Assessments at 3 months were conducted in the participants' homes by bachelor's level research staff masked to allocation.

2.5 | Statistical analyses

Characteristics of the caregivers and the care recipients were compared between the two groups using a two-sample *t* test or chi-square test; the Wilcoxon rank-sum test or Fisher exact test were used if needed due to violations of the underlying assumptions. Primary analysis of the intervention effect used an intent-to-treat approach, in

which groups were compared based on randomization assignment. To account for the clustering of participants by commune, mixed-effects models that included a cluster-specific random effect were used to estimate differences between groups. Models utilized the post-intervention assessment as the outcome and included the baseline assessment as a covariate. The post-intervention assessment was divided by the standard deviation (SD) in the entire sample at baseline, so that interpretation of model coefficients was in terms of standard deviation units. Model assumptions were checked and met by the data. All analyses were conducted in SAS, version 9.4, and a *P*-value $< .05$ was considered statistically significant.

2.6 | Data monitoring, safety, and quality control

The study had a safety officer (SO) but did not have a data safety monitoring board (DSMB). The rationale for not having a DSMB was that this was a single-site, minimal-risk clinical trial with a relatively small number of subjects in total. The lead investigators met with the SO during the clinical trial to review the progress of the study and adverse events.

3 | RESULTS

3.1 | Comparison of subject characteristics in intervention and enhanced control groups

The characteristics of the 51 participants who completed the study are shown in Table 1. In both groups, most caregivers were either daughters or daughters-in-law, and the mean years of formal education were < 9 . Most of the care recipients were female with moderate-to-severe dementia based on the mean CDR scores of 2.6 and 2.5 in the enhanced control group and intervention group, respectively. There were no statistically significant differences in the characteristics at baseline except for the number of hours providing care to the person with dementia each week, which was significantly higher in the enhanced control condition compared with the intervention group ($P = .03$).

3.2 | Primary and second caregiver outcomes

There were no significant baseline differences between the two groups in primary or secondary outcomes, including the ZBI ($P = .9$), PHQ-4 ($P = .5$), and Alzheimer's knowledge ($P = .7$) (see Table 2). Using analysis of covariance (ANCOVA) to evaluate the intervention effect, with the 3-month assessment as the outcome and the baseline assessment as a covariate, we found that the REACH VN group had average ZBI scores 1.2 SD lower ($P = .02$) than the enhanced control group at 3 months. For the secondary outcomes, PHQ-4 scores for the REACH VN group were 0.7 SD ($P = .03$) lower than the enhanced control group at 3 months (see Figure 1). There was no significant difference in change in Alzheimer's knowledge scores (estimated difference = 0.2 SD lower in REACH VN compared to the enhanced control, $P = .7$, 95% confidence interval [CI]

TABLE 1 Pilot cluster RCT caregiver participant characteristics

	Enhanced control (n = 26)	REACH VN (n = 25)	P-value
Caregiver characteristics			
Age ^a	58.7 (13.9)	59.0 (10.4)	.9
Education (y) ^b	7.2 (2.6)	8.3 (3.4)	.2
Male (%)	19.2	8.0	.4
Marital status (%)			.4
Never married	0	0	
Married/living with partner	96.1	88.0	
Separated/divorced	3.8	8.0	
Widowed	0	4.0	
Other	0	0	
Relationship to care recipient (%)			.6
Son/son-in law	7.7	0	
Daughter/daughter-in-law	65.4	68.0	
Spouse	26.9	28.0	
Other	0	4.0	
Caregiving (y) ^c	9.5 (10.6)	7.0 (7.0)	.3
Daily caregiving (%) ^d			.03
< 1 h	0	0	
1-2 h	3.8	12.5	
3-4 h	3.8	25.0	
>4 h	92.3	62.5	
Care recipient characteristics			
Age	82.8 (11.1)	83.7 (11.6)	.8
Male (%)	23.1	24.0	1.0
CDR	2.6 (0.6)	2.5 (0.6)	.2

Abbreviations: CDR, Clinical Dementia Rating; RCT, randomized controlled trial; REACH VN, Resources for Advancing Alzheimer's Caregiver Health in Vietnam.

^aCaregiver age missing for two individuals.

^bCaregiver education missing for five individuals (three Enhanced Control, two REACH VN).

^cCaregiving (years) missing for three individuals (two Enhanced Control, one REACH VN).

^dDaily caregiving missing for one REACH VN.

TABLE 2 Analysis of primary and secondary outcomes

	Enhanced control (n = 26)	REACH VN (n = 25)	P-value*
Primary caregiver outcome			
ZBI (baseline)	6.8 (2.3)	6.7 (2.1)	.9
ZBI (3 mo)	5.5 (3.4)	2.9 (2.4)	.02
Secondary caregiver outcomes			
PHQ-4 (baseline)	4.6 (2.8)	4.1 (3.0)	.5
PHQ-4 (3 mo)	2.9 (2.8)	0.9 (1.0)	.03
AD Knowledge (baseline)	18.8 (2.1)	19.0 (2.2)	.7
AD Knowledge (3 mo)	20.1 (2.0)	19.9 (3.2)	.7

Abbreviations: AD Knowledge, Alzheimer's Disease Knowledge Scale; PHQ-4, Patient Health Questionnaire-4; REACH VN, Resources for Advancing Alzheimer's Caregiver Health in Vietnam; ZBI, Zarit Burden Inventory 4-item.

*P-value is from a model accounting for the clustering of individuals in the same community. Models for the 3-month outcomes included the baseline value as a covariate.

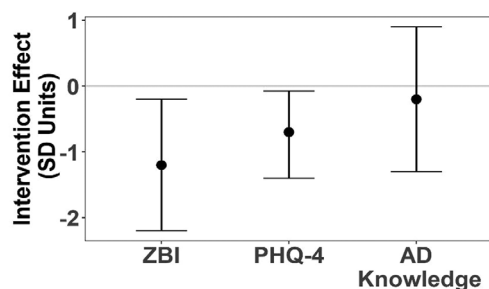


FIGURE 1 Effect size for primary and secondary caregiver outcomes in a cluster randomized controlled trial (RCT)

for difference: $-1.3, 0.9$). Secondary analyses also considered the addition of caregiver hours to the model, as this was significantly different between the groups, but there was no impact on the effect sizes (ZBI: 1.1 SD; PHQ-4: 0.7 SD). No adverse events occurred among study participants (ie, family caregivers).

3.3 | Feasibility outcomes

A total of 148 caregivers in 12 clusters were screened for eligibility (see Figure 2). Three (25%) of the 12 clusters (one randomized to the intervention and two to the enhanced control condition) were excluded due to low recruitment (fewer than three eligible caregivers identified). Of the 146 caregivers who were screened in the remaining nine clusters, 71 met study criteria, and 60 (85% of those caregivers who were eligible) were successfully randomized. Of the 71 eligible participants, 7 (10%) declined to participate and an additional 4 were excluded because the target of 60 caregivers had been reached. Of the 60 participants randomized (across nine clusters), 51 completed the study (26 in the control group and 25 in REACH VN intervention). Reasons for caregiver attrition included care recipient passing away ($n = 5$, two in intervention and three in control), caregiver decided not to participate ($n = 3$, due to lack of time and other reasons), and one caregiver was found to be ineligible due to significant cognitive impairment. Among the 25 caregivers who completed the intervention, 100% completed at least three sessions. Based on a review of standardized treatment delivery forms completed by the interventionists at the end of each session for the REACH VN groups, all required elements were completed for 98% of session 1, 96% for session 2, 92% for session 3, 87% for session 4, and 100% of final (closure) sessions. Baseline and 3-month assessments were successfully completed for all 51 participants who completed the study.

4 | DISCUSSION

To our knowledge this is the first study to test the efficacy of an AD/ABD family caregiving intervention in Vietnam and one of the few conducted in an LMIC in Asia.⁴ The REACH VN intervention, a culturally adapted caregiver intervention, was both efficacious and feasible

when delivered to a community-based sample of caregivers in a semi-rural area outside of Hanoi, Vietnam.

For our primary outcome (caregiver burden), family caregivers who were in the intervention group had improved outcomes compared with those in the enhanced control group. Despite significant improvement in burden from baseline to 3 months in our enhanced control group, the effect size demonstrated in this study is considerably larger compared with effect sizes for REACH VA and other comparable interventions in the United States. The effect sizes for caregiver depression and anxiety symptoms, while not as large as for the primary outcome, were also larger than most caregiver studies conducted in the United States.⁸ One possible reason for the increased effect sizes in Vietnam may be attributable to the relative lack of knowledge, supports, and services available to family caregivers in Vietnam compared with the United States.

Although we also hypothesized that AD knowledge gain would be larger in the intervention group, both groups experienced a very modest knowledge gain, and there were no significant difference between groups. Our findings differ from studies conducted in the United States, which have found significant improvements in Alzheimer's knowledge as a result of multi-component interventions like REACH VN.⁸ One possible explanation for the lack of between group differences is that participants in both the intervention and enhanced control received education about AD. However, this does not explain the very modest gain in knowledge for both groups. The measure we used to assess Alzheimer's knowledge has been used in English and Vietnamese populations residing outside Vietnam who had higher levels of formal education compared with the caregivers in our study (mean of 7-8 years of formal education). The measure may need to be modified to be more suitable for caregivers with a lower level of formal education. Finally, the knowledge gained by caregivers (eg, skills in managing dementia-related behaviors, communicating with the person with dementia, and managing their own distress) may not be captured by the Alzheimer's knowledge scale, which focuses exclusively on the nature of the disease.

The intervention was feasible as well as efficacious. Our retention in the trial was 85%, and most of attrition was due to high mortality rates among the care recipients in the trial. The mortality rates in this study are comparable with another study conducted in India²⁷ and may be attributable to the advanced age of the care recipients, most of whom were living with moderate-to-severe dementia. At the cluster level, three clusters were dropped due to lack of insufficient recruitment. The reasons for the low recruitment in these clusters is less clear and may reflect lower levels of local interest, resources, and capacity to assist the research team in the recruitment process.

The interpretation of our results should consider several study limitations. The pilot cluster RCT was conducted in a semi-rural area in northern Vietnam, and caution should be used in generalizing these results to other settings, particularly urban areas, where the social and economic circumstances of older adults and family caregivers may be quite different. Inclusion of a knowledge scale that may not have been suitable for this population's lower literacy level is a second limitation of this study. Finally, our study is an

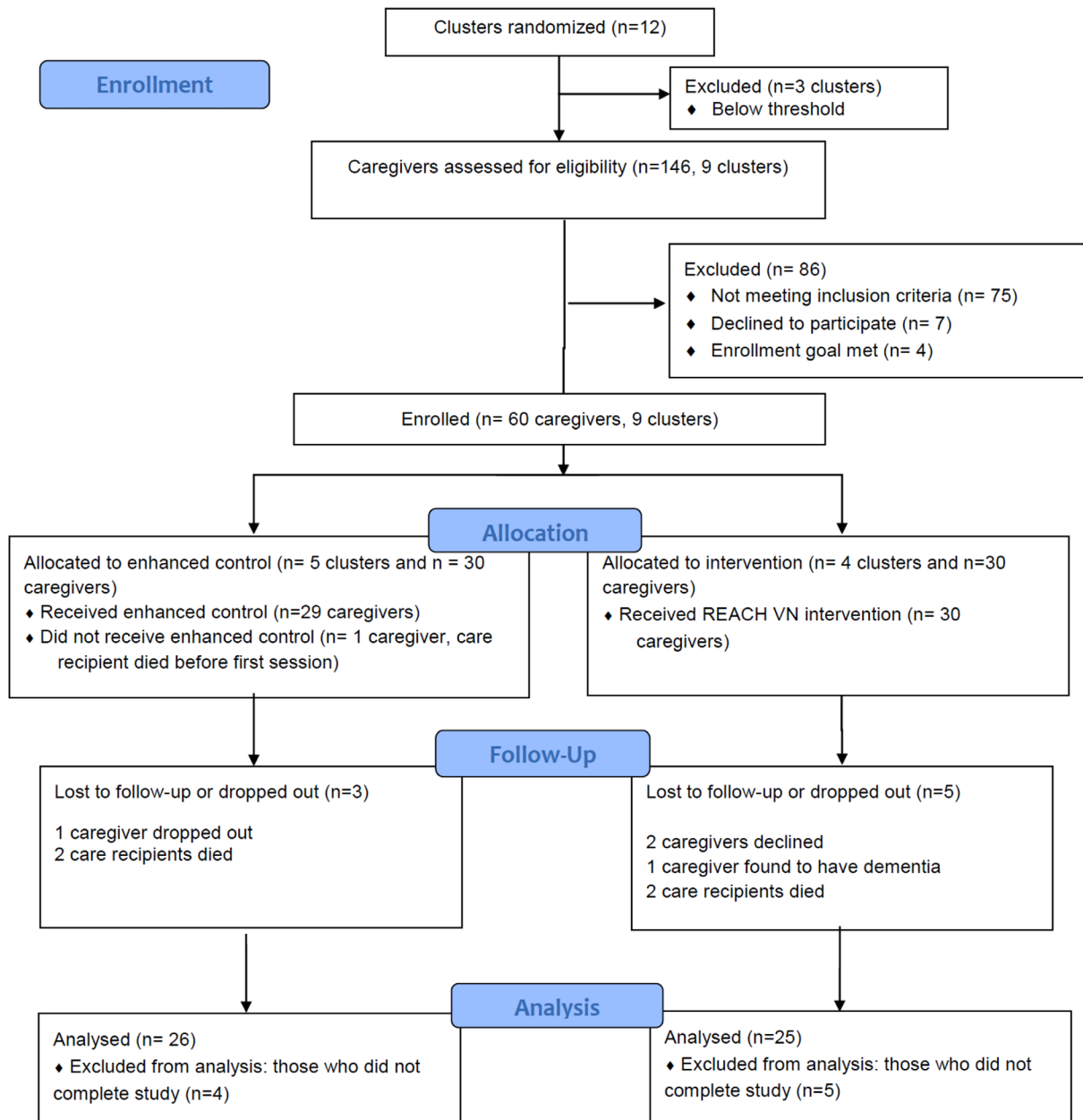


FIGURE 2 Consort diagram of recruitment for pilot cluster randomized controlled trial (RCT)

efficacy study in which the interventionists were highly trained social workers, doctors, and nurses from the Vietnam National Geriatric Hospital. To move toward a more scalable intervention model that is sustainable, future research is needed to determine if providers at the local level can be trained to deliver the intervention with fidelity and to achieve similar outcomes. Our study did not measure a number of variables that might be important to include in future studies in relationship to caregiver outcomes, including caregiver religion and socioeconomic status, as well as dementia type, medical comorbidities, and health insurance of the person living with dementia.

In conclusion, we found that a culturally adapted and relatively brief family caregiver intervention was both efficacious and feasible in Vietnam. These results may be relevant to other LMIC countries in

Asia that are facing growing numbers of older adults with dementia in the face of relatively little evidence about what community programs are effective. Our study suggests that with relatively modest cultural adaptation, multicomponent interventions developed in the United States may be effective for LMICs in Asia and perhaps elsewhere. Evidence already exists for the efficacy of multi-component interventions based on REACH and other models in HIC in Asia.⁴ Furthermore, REACH VN's relative brevity may enhance its sustainability in other lower resource settings. Although our findings clearly need replication in a larger study with more community-embeddedness, these results are promising for Vietnam and other LMICs with a pressing need to meet the challenge of providing community-based care and support to family caregivers of people living with dementia.

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CONFLICT OF INTEREST

The authors have no conflicts of interest to disclose.

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ClinicalTrials.gov Search Results 12/07/2022

	NCT Number	Title	Other Names	Status	Conditions	Interventions	Characteristics	Population	Sponsor/ Collaborators	Funder Type	Dates	Locations
1	NCT03587974	Randomized Controlled Trial to Test an Alzheimer's Family Caregiver Intervention in Vietnam Study Documents: Informed Consent Form Statistical Analysis Plan Study Protocol	Title Acronym: Other Ids: 1223433	Completed	•Family Caregivers	•Behavioral: REACH-VIETNAM	Study Type: Interventional Phase: Not Applicable Study Design: •Allocation: Randomized •Intervention Model: Parallel Assignment •Masking: Single (Outcomes Assessor) •Primary Purpose: Supportive Care Outcome Measures: •Zarit Burden Inventory-4 •Patient Health Questionnaire-4 •Alzheimer's Disease Knowledge Scale	Enrollment: 60 Age: 18 Years and older (Adult, Older Adult) Sex: All	•University of California, Davis	•Other	Study Start: June 10, 2018 Primary Completion: October 31, 2018 Study Completion: October 31, 2018 First Posted: July 16, 2018 Results First Posted: July 22, 2020 Last Update Posted: July 22, 2020	•Vietnam National Geriatric Hospital, Hanoi, Vietnam



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RESEARCH

ASSESSMENT OF THE IMPACT OF DEMENTIA CARE AND SUPPORT PROGRAM IN BOTH PATIENT AND CAREGIVER OUTCOMES: AN INTERVENTION STUDY

ABSTRACT

Introduction: Dementia is one of the priority issues among the public health concerns. This study aims to assess the impact of the Dementia Care and Support Program in caregivers and patients with dementia.

Materials and Methods: This study is an intervention study. Dementia Care and Support Program was a planned 16-week program. Sixty-one patient-caregiver pairs were randomized into two groups, the intervention (n=31) and control groups (n=30). Dementia Care and Support Program was applied to the intervention group, but the control group received routine hospital care. Data were collected between July and November 2016 from a dementia outpatient clinic. While data for patients were collected using the quality-of-life assessment in Alzheimer's disease and Neuropsychiatric Inventory, data for caregivers were collected using the quality of life SF-36, Beck Depression, Beck Anxiety, and Zarit Caregiver Burden Care Inventory.

Results: Fifty-four patient-caregiver pairs completed the study. The mean age of the patients was 76.7±11.2 (46-96) years old. There was no statistically significant difference in the quality-of-life scores and neuropsychiatric symptom scores between the patient groups (p>0.05). The mean age of caregivers was 53.6±14.8 (22-81) years old. Statistically significant differences were found in NPI-D, quality-of-life mental health, quality-of-life physical health, depression, and anxiety scores between the caregiver groups (p<0.05), but there was no statistically significant difference in burden scores (p>0.05).

Conclusion: This study established that Dementia Care and Support Program has positive effects on caregivers.

Keywords: Dementia; Patient; Caregiver; Quality of Life; Depression.

ARAŞTIRMA

DEMANS BAKIM VE DESTEK PROGRAMININ HASTA VE BAKIM VERİCİ ÇIKTILARI ÜZERİNDEKİ ETKİSİNİN DEĞERLENDİRİLMESİ: BİR MÜDAHALE ÇALIŞMASI

Giriş: Demans halk sağlığı sorunları arasında öncelikli bir konudur. Bu çalışmanın amacı demanslı hastaların bakım vericilerine uygulanan Demans Bakım ve Destek Programı'nın etkisini değerlendirmektir.

Gereç ve Yöntem: Bu çalışma 16 haftalık bir müdahale çalışmasıdır. Çalışmada 61 hasta-bakım verici çifti iki gruba randomize edildi. Girişim grubu 31, kontrol grubu 30 kişiden oluştu. Girişim grubuna Demans Bakım ve Destek Programı uygulanırken, kontrol grubuna rutin hastane bakımı verildi. Veriler Demans polikliniğinde 2016 Temmuz-2016 Kasım tarihleri arasında toplandı. Veri toplamada hastalar için Alzheimer Hastalığı Yaşam Kalitesi Ölçeği ve Nöropsikiyatrik Envanter kullanılırken; bakım vericiler için Yaşam Kalitesi SF 36 Ölçeği, Beck Depresyon Ölçeği, Beck Anksiyete Ölçeği, Zarit Bakım Yüklü Ölçeği kullanıldı.

Bulgular: Çalışma 54 bakım verici-hasta çifti ile sonlandırıldı. Hastaların yaş ortalaması 76.7±11.2 (46-96) idi. Hastaların yaşam kalitesi puanlarında ve nöropsikiyatrik semptom ciddiyeti (NPI-S) puanlarında gruplar arasında istatistiksel olarak anlamlı bir fark bulunmadı (p>0.05). Bakım vericilerin yaş ortalaması ise 53.6±14.8 (22-81) idi. Bakım vericilerin distress (NPI-D) puanlarında, yaşam kalitesi puanlarında, depresyon ve anksiyete puanlarında gruplar arasında istatistiksel olarak anlamlı fark bulundu (p<0.05). Fakat, bakım verici yükünde gruplar arasında istatistiksel olarak anlamlı bir fark bulunmadı (p> 0.05).

Sonuç: Bu çalışma, Demans Bakım ve Destek Programı'nın demans hastalarının bakım vericileri üzerinde olumlu etkisi olduğunu ortaya koymaktadır.

Anahtar Sözcükler: Demans; Hasta; Bakım Verici; Yaşam Kalitesi; Depresyon.



INTRODUCTION

Dementia is a rapidly growing global public health problem. Approximately 50 million people have dementia worldwide, and most of them live in low- and middle-income countries. Nearly 10 million new cases are discovered annually (1). Dementia rises exponentially during old age; it is one of the leading causes of disability and dependency among the elderly (2). Dementia is often seriously devastating for both patient and caregiver or family. Family caregivers who are informal caregivers experience burden, depression, anxiety, health-related problems, and financial difficulties (1,2).

Dementia care aims to preserve patients' functioning, decrease disability, regulate the environment and relationships to sustain stability, and maintain personality and quality of life (3). Nurses are accountable for providing education, information, and support to caregivers during all stages, which is the first and fundamental step among the non-pharmacological approaches (3, 4). Based on evidences, public health nursing practices in dementia care aim to maintain quality results despite differences in patient care outcomes, standardized care, and enhanced nurse satisfaction (5).

The program used in this research was Dementia Care and Support Program (DCSP), which was developed by the authors from the literature review (6, 7). This was a multicomponent psychosocial intervention. The psychosocial intervention for family and/or informal caregiver attempts to increase knowledge (education), improve skills and coping strategies, and provide support (7). In the recent years, several authors have assessed the efficacy of interventions for both caregivers and dementia patient, but they have found considerable variability in the outcome. The DCSP consists of both individual (telephone interviews and home visits) and group support intervention (group training, face-to-face group meeting, WhatsApp group interviews). When the literature was examined, only two nursing

researches (8, 9) were found in Turkey. This situation has increased the interest in conducting research about the subject. This study aims to provide evidence for the impact of multicomponent and multidisciplinary interventions on both patient and caregiver outcomes.

MATERIALS AND METHOD

Study design

This study is intervention study.

Participants

This study was conducted at Manisa Celal Bayar University (MCBU) Hafsa Sultan Hospital Dementia Outpatient Clinic in Manisa, between July and November 2016. The study universe comprised clinic-registered patients (N = 396) and their caregivers (family caregivers). We computed the sample size with power analysis, statistically based on the work of Martin-Corasso et al. (10). The power analysis for this study, at 99% power and 0.01 error level, was therefore determined that 38 patient-caregiver pairs (19 patient-caregiver pairs in each group) were required. Considering loss of subjects over time, we initiated the study with 61 patient-caregiver pairs. The study sample was selected by probabilistic sampling among patient-caregiver pairs that fulfilled the inclusion criteria.

The inclusion criteria for patients were as follows: clinical diagnosis of dementia; living in Manisa; not leaving Manisa during the study period and willing to participate in the research. The exclusion criterion was diagnosis with cancer. We enrolled caregivers based on the following criteria: aged ≥18 years; living in Manisa; not leaving Manisa during the study period and willing to participate in the research.

Randomisation criteria were dementia type, dementia stage and caregivers' education level. We stratified the patients and caregivers based on the education status of caregivers, considering

the randomisation criteria. Using simple random placement method, we determined groups of caregivers to be assigned to the groups. The patient-caregiver pairs were assigned to the randomised IG (n = 30) and CG (n = 31). We evaluated the homogeneity of randomisation using the χ^2 test. Figure 1 shows the study flowchart.

Procedures

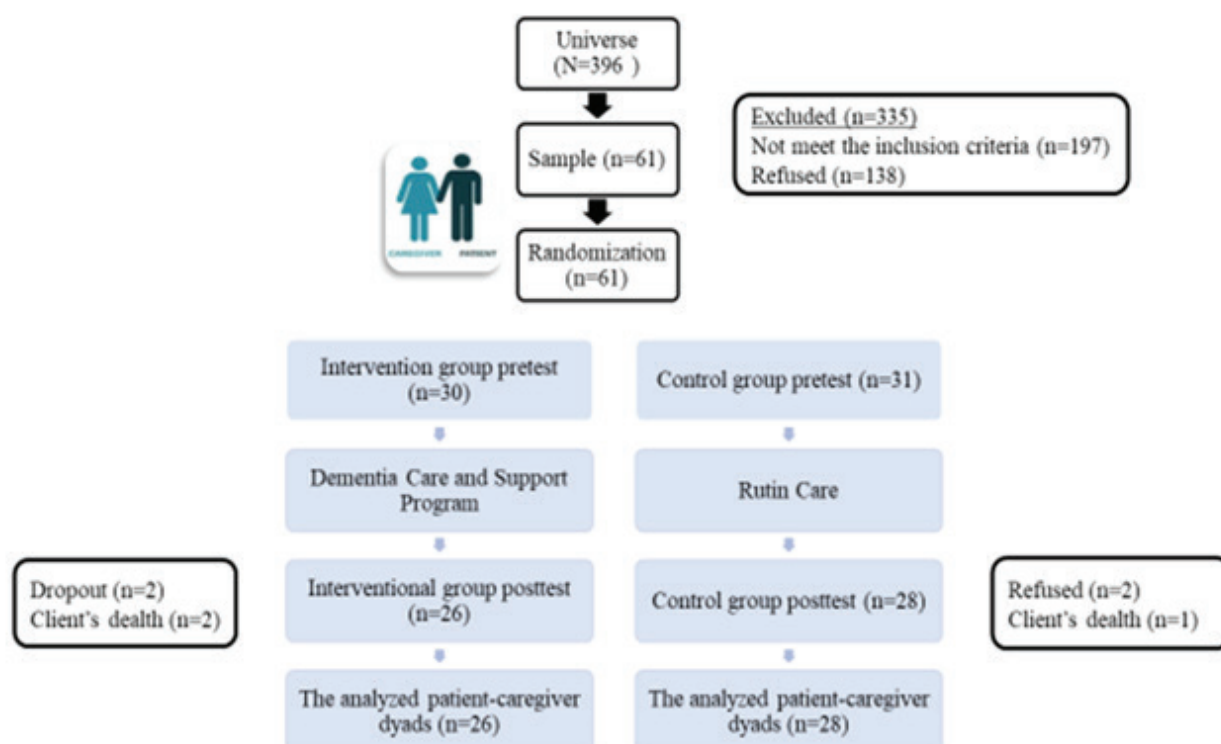
The programme with multi-component interventions was given by a multi-disciplinary team (nurse, neurologist, psychologist). Dementia type and its stages were diagnosed by a neurologist.

Intervention group

The group received group training and support, home visits and individual support through telephonic interviews. Caregivers were unable to attend group training when they could not find anyone to leave their patients. This handicap was tried to be overcome with the WhatsApp group and house visits. WhatsApp group was set up to make caregivers communicate and interact with each other and researchers. During 16 weeks, caregivers were interviewed routinely per month; four face-to-face interviews were conducted with each patient-caregiver dyad.

After pre-test, intervention group was trained with a structured standard training package

Figure 1. Flowchart of the Research.





(Fig. 2); the training comprised videos, oral presentation, question–answer sessions. Training programme created by the researchers and the study plan were explained to the caregivers in writing. In some training sessions, psychologist at the dementia clinic played a supporting role (stress management and problem-solving methods). Furthermore, caregivers provided support for individual problems.

Control group

We interviewed the CG ($n = 31$) two times during the data collection process. The group received just routine hospital care. At the end of 16 weeks, the programme was provided to this group too.

Data collection

Caregiver and Patient Information Form: The form prepared by referring to the literature (10, 11). was comprised of two sections. Firstly section was about patients' age, sex, education level, marital status, dementia type and stage, comorbidity etc. Secondly section was about caregivers' age, sex, education level, marital status, income status, relationship with patients etc.

Patients Quality-of-life in Alzheimer's Disease (QoL-AD): The scale developed by Longsdon et al. (1999) was filled and assessed separately for patients and caregivers. The scale measures quality of life of Alzheimer's patients. The scale comprises 13 items of the Likert type, and quality of life increases as the score increases. Akpınar & Küçükçüçlü (2012) assessed the Turkish reliability and validity of the and Cronbach's α to be 0.84 (12). In this study, Cronbach's α was 0.84.

Neuropsychiatric Inventory (NPI): This assesses the frequency and severity of psychological and behavioural symptoms of dementia and is based on interviews with caregivers. It comprises two parts as follows: (a) measures the severity of associated problem behaviours on a scale of 1–3 (NPI-S) and (b) measures the perceived distress of problem behaviours by caregivers on a scale of 0–5 (NPI-D).

Akça-Kalem et al. (2005) evaluated Cronbach's α of the inventory to be 0.79. For specific items of the frequency and severity, Cronbach's α varies between 0.76 and 0.79, respectively (13). In this study, Cronbach's α of the inventory was 0.91, and for specific items of the frequency and severity Cronbach's α values were 0.73 and 0.77.

Caregivers

Quality of Life Scale SF36 comprises eight sub-dimensions and defines two summary areas: mental health and physical health. The total score can be 0–100. Koçyiğit et al. (1999) assessed the Turkish reliability and validity of the scale. Cronbach's α of the scale was 0.73–0.76 for sub-dimensions of the scale (14); in this study, it was 0.73–0.88.

The Beck Depression Inventory (BDI) was developed by Beck et al. (1961) and enables the assessment of both depressive symptoms and cognitive content. The total score can be 0–63 points. Hisli (1989) assessed the Turkish validity and reliability. The total score can be 0–63 points. Cronbach's α was 0.80 (15); in this study, it was 0.80.

Beck Anxiety Inventory (BAI) is a 21-item Likert-type scale, developed by Beck et al. (1988), to determine the frequency of anxiety symptoms of individuals. The severity of anxiety increases as the total score of the scale increases. The total score can be 0–63 points. Ulusoy (1998) established the Turkish validity and reliability of the scale. Cronbach's α of the scale was 0.93 (16); in this study, it was 0.87.

Zarit Burden Interview (ZBI) the most widely used in caregiver burden. It encompasses the physical, emotional and financial burden as perceived by the caregiver. It asks 22 questions that are rated on a five-point scale (0 = not at all to 4 = nearly always). Cronbach's α for the validity study of the scale was 0.95 (17); in this study, it was 0.91.

Statistical analysis

Statistical analyses were performed using

the SPSS 15.0 software package. We evaluated descriptive characteristics using numbers and percentages and binary comparisons using the χ^2 test. We considered the statistical significance at 0.05. Data of the normal distribution are presented by the number of sample size and Shapiro–Wilk values. Furthermore, non-parametric tests were used to analyse non-normally distributed variables.

Ethics

The study was approved by the MCBU Medicine Faculty Local Ethics Board (dated 9 December 2015; No: 20478486-412), MCBU Hafsa Sultan

Hospital. We obtained written informed consent from all caregivers and some of the patients through the Volunteer Form.

RESULTS

We assessed the research findings in two parts – patients' and caregivers' results.

Patients' results

The mean age of the patients was 76.7 ± 11.2 (46–96) years old. We observed no statistically significant differences between the groups for all demographic characteristics ($p > 0.05$), except for marital status ($p < 0.05$; Table 1).

Table 1. Socio-demographic and descriptive characteristics of patients (n = 61)

Variable		Intervention Group (n = 30)	Control Group (n = 31)	χ^2 test*
		n (%)	n (%)	χ^2 , p
Age	≤76 years	9 (40.9)	13 (59.1)	$\chi^2 = 0.942$ p = 0.332
	>76 years	21 (53.8)	18 (46.2)	
Sex	Female	23 (52.3)	21 (47.7)	$\chi^2 = 0.604$ p = 0.437
	Male	7 (41.2)	10 (58.8)	
Education Level	Illiterate	8 (66.7)	4 (33.3)	$\chi^2 = 8.019$ p = 0.155
	Literate	3 (25.0)	9 (75.0)	
	Primary School	15 (62.5)	9 (37.5)	
	Middle School	1 (25.0)	3 (75.0)	
	High School	2 (40.0)	3 (60.0)	
	University	1 (25.0)	3 (75.0)	
Marital Status	Married	10 (30.3)	23 (69.7)	$\chi^2 = 10.250$ p = 0.001
	Single	20 (71.4)	8 (28.6)	



Variable		Intervention Group (n = 30)	Control Group (n = 31)	χ^2 test*
		n (%)	n (%)	χ^2, p
Dementi Type^a	Alzheimer's Disease	23 (46.0)	27 (54.0)	$\chi^2 = 2.104$ $p = 0.717$
	Typea	1 (50.0)	1 (50.0)	
	Dementia with Lewy Bodies	1 (50.0)	1 (50.0)	
	Mixed Dementia	1 (50.0)	1 (50.0)	
	Parkinson's Disease Dementia	4 (80.0)	1 (80.0)	
Dementia Stage^a	Very Mild (0.5)	2 (50.0)	2 (50.0)	$\chi^2 = 3.710$ $p = 0.295$
	Dementia Stagea	11 (47.8)	12 (52.2)	
	Moderate (2)	12 (66.7)	6 (33.3)	
	Severe (3)	5 (33.3)	10 (66.7)	
Comorbidity	Yes	21 (47.7)	23 (52.3)	$\chi^2 = 0.133$ $P = 0.715$
	Comorbidity	9 (52.9)	8 (47.1)	

Abbreviations: * χ^2 test; a Randomisation criteria.

There was no statistically significant difference in quality-of-life scores and neuropsychiatric symptom scores between the groups ($p > 0.05$; Table 2).

Caregivers' results

The mean age of caregivers was 53.6 ± 14.8 (22–81) years old. No significant differences were observed between the groups for all demographic characteristics ($p > 0.05$; Table 3).

The results of the present study demonstrate that there was statistically significant differences in NPI-D, quality-of-life (both mental and physical health), depression, and anxiety scores between the groups ($p < 0.05$), but there was no statistically

significant difference in burden scores ($p > 0.05$; Table 4).

DISCUSSION

Caregivers have critical roles, needs, and difficulties; however, inadequate information about the care provided by the caregivers increases problems for patients as well as caregivers. With the use of structured sessions and information and psychological support investigation, one of the hopes of the 16-week intervention program was to determine whether intervention could affect the outcomes for both caregivers and dementia patients.

Table 2. Dependent variable scores at pre-test–post-test measures of patients with dementia (n = 61).

		Intervention Group		Control Group		Between groups**
		n	$\bar{x} \pm SD$ (Min; Max)	n	$\bar{x} \pm SD$ (Min; Max)	
QoL-ADa	Pre-test	22	31.18 $\pm$ 6.14 (22; 42)	20	33.29 $\pm$ 4.96 (22; 44)	$z = -2.054$; $p = 0.400$
	Post-test	17	31.79 $\pm$ 5.41 (22; 39)	17	32.39 $\pm$ 4.33 (22; 39)	$z = -0.328$; $p = 0.743$
NPI-S	Pre-test	30	31.18 $\pm$ 23.79 (0; 87)	31	19.41 $\pm$ 16.09 (0; 52)	$z = -0.845$; $p = 0.398$
	Post-test	26	26.50 $\pm$ 21.99 (0; 77)	28	30.70 $\pm$ 23.78 (8; 88)	$z = -1.013$; $p = 0.311$

Abbreviations: *Mann–Whitney U-test.; **Wilcoxon marked rank test.; a Calculated solely for Alzheimer's disease; QoL-AD, Quality of life in Alzheimer's Disease; NPI-S, Neuropsychiatric Inventory –Severity.

Table 3. Socio-demographic and descriptive characteristics of caregivers (n = 61).

Variable		Intervention Group (n = 30)	Control Group (n = 31)	χ^2 test*
		n (%)	n (%)	χ^2 , p
Age	≤53 years	18 (58.1)	13 (41.9)	$\chi^2 = 1.991$; $p = 0.158$
	>53 years	12 (40.0)	18 (60.0)	
Gender	Female	25 (52.1)	23 (47.9)	$\chi^2 = 0.759$; $p = 0.384$
	Male	5 (38.5)	8 (61.5)	
Education Level	Illiterate	–(–)	1 (100.0)	$\chi^2 = 0.984$; $p = 0.964$
	Literate	2 (50.0)	2 (50.0)	
	Primary School	10 (50.0)	10 (50.0)	
	Middle School	3 (50.0)	3 (50.0)	
	High School	9 (50.0)	9 (50.0)	
	University	6 (50.0)	6 (50.0)	
Marital Status	Married	23 (50.0)	23 (50.0)	$\chi^2 = 0.050$; $p = 0.823$
	Single	7 (46.7)	8 (53.3)	
Income Status	Low Income	4 (30.8)	9 (69.2)	$\chi^2 = 2.689$; $p = 0.261$
	Middle Income	24 (55.8)	19 (44.2)	
	High Income	2 (40.0)	3 (60.0)	



Variable		Intervention Group (n = 30)	Control Group (n = 31)	χ^2 test*
		n (%)	n (%)	χ^2, p
Relationship with Patients	Spouse	4 (25.0)	12 (75.0)	$\chi^2 = 10.211$; $p = 0.069$
	Child	19 (63.3)	11 (36.7)	
	Daughter-in-law	6 (54.5)	5 (45.5)	
	Sibling	1 (100.0)	-(-)	
	Parents	-(-)	1 (100.0)	
	Other	-(-)	2 (100.0)	

Abbreviations: * χ^2 test; a Randomisation criteria.

Table 4. Dependent variable scores at pre-test–post-test measures of caregivers (n = 61).

		Intervention Group		Control Group		Between groups**
		n	$\bar{x} \pm SD$ (Min; Max)	n	$\bar{x} \pm SD$ (Min; Max)	
SF 36 (physical health)	Pre-test	30	41.75 $\pm$ 10.00 (26; 52)	31	41.59 $\pm$ 9.78 (25; 61)	$z = 0.447$; $p = 0.655$
	Post-test	26	45.41 $\pm$ 7.94 (30; 57)	28	37.74 $\pm$ 10.77 (22; 55)	$z = 2.372$; $p = 0.018$
SF36 (mental health)	Pre-test	30	41.79 $\pm$ 9.024 (17; 55)	31	42.18 $\pm$ 10.52 (24; 57)	$z = 0.0736$; $p = 0.462$
	Post-test	26	44.87 $\pm$ 8.94 (21; 60)	28	39.47 $\pm$ 9.39 (27; 57)	$z = 2.701$; $p = 0.007$
BDI	Pre-test	30	11.62 $\pm$ 8.34 (1; 35)	31	15.23 $\pm$ 7.70 (3; 26)	$z = 1.525$; $p = 0.127$
	Post-test	26	9.68 $\pm$ 8.45 (0; 35)	28	20.64 $\pm$ 8.94 (5; 39)	$z = 3.268$; $p = 0.001$
BAI	Pre-test	30	10.06 $\pm$ 8.89 (0; 30)	31	12.23 $\pm$ 8.05 (1; 30)	$z = 0.043$; $p = 0.965$
	Post-test	26	6.18 $\pm$ 7.26 (0; 29)	28	15.47 $\pm$ 9.28 (1; 34)	$z = 2.636$; $p = 0.008$
NPI-D	Pre-test	30	11.87 $\pm$ 9.73 (0; 32)	31	9.35 $\pm$ 8.06 (0; 27)	$z = -0.296$; $p = 0.767$
	Post-test	26	10.50 $\pm$ 8.70 (0; 27)	28	14.64 $\pm$ 10.06 (3; 38)	$z = -1.968$; $p = 0.049$
ZBI	Pre-test	30	31.43 $\pm$ 18.18 (5; 74)	31	40.17 $\pm$ 18.10 (13; 65)	$z = 0.599$; $p = 0.549$
		26	27.87 $\pm$ 18.39 (5; 63)	28	40.70 $\pm$ 17.66 (10; 70)	$z = 1.351$; $p = 0.177$

Abbreviations: *Mann–Whitney U-test; **Wilcoxon marked rank test, SF 36, Quality of Life Scale SF36; BDI, Beck Depression Inventory; BAI, Beck Anxiety Inventory; NPI-D, Neuropsychiatric Inventory-Distress, ZBI; Zarit Burden Interview..

Figure 2. The Dementia Care and Support Programme's structured standard training package.

Session	Duration	Theme
Session 1	30 min.	Obtaining information about the patient and the relatives responsible for the care
Session 2	40 min.	What is dementia?
Session 3	30 min.	Dementia treatment (pharmacologic and on-pharmacological treatments)
Session 4	40 min.	Dementia patient care (nutrition, communication, incontinence, sleep, pressure scar...
Session 5	30 min.	Importance of family support in the treatment and care of the patient
Session 6	30 min.	Problem solving methods
Session 7	30 min.	Share experience of stress with patients
Session 8	30 min.	Importance of communication in the recovery of patients

Patients

The quality of life is an essential parameter in terms of the progression of dementia. Thus, quality-of-life assessment is imperative in planning the care given to the patients. In this study, there was no statistically significant difference in the quality of life between the groups. In the literature, studies on the quality of life of dementia patients are limited. Soylemez et al. conducted an intervention study with a 6-month period while Koivisto et al. made a 3-year follow-up (8, 18). In both studies, no statistically significant difference between the groups in the quality of life (8, 18) was observed. Our findings on the quality of life may be due to the patients' lack of insight and cognitive deficits.

In our study, the intervention provided no beneficial effects on patients' NPI-S score. An 18-month follow-up study provided beneficial results on NPI (11). Our findings are consistent studies both with short intervention period (8, 19) and with long intervention period (18). The findings of some studies and our study (8, 18) supported the idea that neuropsychiatric problems are associated with lower quality of life in patients.

Caregivers

We observed a marked difference between

the groups in the post-test on both areas of the caregivers' quality-of-life scores. These findings are consistent with some studies when the psychosocial intervention was applied (10, 11). This finding reinforces that modifying the intervention according to the caregiver's needs may result to a successful outcome.

Reportedly, caregivers of patients with dementia are hidden victims who experience mental and physical health-related problems that hinder their ability to cope with stress effectively (20). In this study, the DCSP has a positive effect on depression in the intervention group. This finding is consistent with the study of Kuo et al. (21) while not consistent with the study of Soylemez et al. (8). Our positive findings may have been due to the efficacy of the multicomponent intervention (group training and support, home visits, and individual support through telephone interviews).

This study shows that psychosocial interventions may also decrease the anxiety of caregivers. Akkerman and Ostwald reported that caregivers exhibited a considerable reduction in anxiety scores (22). Santos et al. emphasized the impact of participating in psychosocial interviews, support groups, and education in decreasing the adverse impact of caregiving (23). This result may have been due to the efficacy of having



an individual consultant and support through telephone interviews. Outside of the initiative program, caregivers were able to reach out to the researcher when they needed it.

In this study, significant difference was observed between the groups' NPI-D post-test scores. Reportedly, neuropsychiatric problems are associated with depression symptoms and caregivers' burden (23). Consistent with Dias's findings, the psychosocial intervention positively affected the caregiver distress (19). Our results suggest that providing education and counseling to caregivers decrease their distress.

Upon evaluating the caregivers' burden scores, we did not observe a marked difference between the groups. These findings were consistent with the findings of some previous studies (23, 24). Martin-Corasso et al. continued intervention for 10 months and showed better burden results (10). Caregiver burden encompasses the physical and psychological well-being, social life, and financial status. In this study, it was thought that this resulted to caregivers' physical and psychological well-being and social life positive progress. However, the lack of intervention in the financial dimension of the burden of care in the study may be the reason for no difference between the groups.

The factor that differentiates this research from other studies in the literature and provides a positive effect on many aspects of the research may be related to the fact that the program applied to caregivers is supported not only by the group trainings given in the hospital but also by the home visits and telephone conversations provided outside of the clinical setting as well as by having a multidisciplinary team.

This study has several limitations. First, no scale was used to determine the patient's quality of life for the different types of dementia, and data were collected by the scale used only on patients with Alzheimer's disease. Second, the study duration and recurrence measurements were limited. There

could be an improvement in most outcomes as the post-test was performed immediately after the 16-week trial. Thus, had the study period been extended, the results could have been different. Finally, some caregivers could not attend the group training because they could not leave their patients alone at home. This resulted in higher home visits and telephone interactions than planned; this intervention also limited the interaction of caregivers with each other.

In conclusion; this study reveals that multicomponent and multidisciplinary interventions, education, and counseling programs have positive impact on caregivers' distress, quality of life, depression, and anxiety. We think that the results of the patients' quality of life and neuropsychiatric symptom severity are clinically but not statistically significant. This is a positive development for a disease without cure. In addition, regular home visits could potentially improve patients' and caregivers' outcomes. Thus, this study recommends using the program for nurses working with dementia patients and their caregivers. Regular and interdisciplinary training and counseling programs should be organized for caregivers, and their efficacy should be assessed. Furthermore, this study highlights the need to add dementia care education to the curriculum for nurses, based on the application of the program to home healthcare providers. Overall, the use of this program for regular home visits by nurses working in primary care units is recommended.

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Effect of Extended Nursing on the Behavioral and Psychological Symptoms and Cognitive Dysfunction of Patients with Moderate and Severe Alzheimer's Disease

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Chen *et al.*: Effect of Extended Nursing on Moderate and Severe Alzheimer's Disease Patients

The purpose of this investigation is to study the effect of extended nursing on the behavioral and psychological symptoms and cognitive dysfunction of patients with moderate and severe Alzheimer's disease. A total of 102 patients with moderate and severe Alzheimer's disease admitted to the People's Hospital of Nanjing Medical University from February 2016 to March 2018 were randomly divided into the observation group and the control group. The control group received routine health education and nursing

guidance after discharge, while the observation group received additional extended nursing scheme for 1 y, including daily living ability training, body function training, language training, memory training, music therapy, psychological nursing support and offline collective activities of patients. The scores of behavioral pathology assessment scale of Alzheimer's disease, Montreal cognitive assessment, and Barthel activities of daily living were compared between the two groups. There was no significant difference in the scores of behavioral pathology assessment scale of Alzheimer's, Montreal cognitive assessment and Barthel activities of daily living between the two groups on admission. One year later, the behavioral pathology assessment scale of Alzheimer's disease scores of the observation group were significantly lower than that of the control group ($p < 0.05$), the Montreal cognitive assessment scores and Barthel activities of daily living scores of the observation group were significantly higher than that of the control group ($p < 0.05$). The extended nursing scheme of this study improved the behavioral and psychological symptoms of Alzheimer's disease patients significantly, including anxiety, fear, hallucination, delusion, emotional disorder, aggressive behavior, behavioral disorder and circadian rhythm disorder. It could also improve the cognitive ability of Alzheimer's disease patients significantly, and improve their ability to handle daily life activities.

Key words: Alzheimer's disease, extended nursing, behavioral and psychological symptoms, cognitive function, ability of daily life activities

Alzheimer's disease (AD) was a kind of neurodegenerative disorder prone to the elderly, with pathological changes in the nervous system such as the widened sulcus gyri, diffuse atrophy of cerebral cortex and neurofibrillary tangles^[1]. It can affect the health of the patients and social stability seriously. Some neurological dysfunctions such as the irreversible, progressive memory impairment, disability, dementia, aphasia and blindness were found in AD patients^[2]. Different from the vascular dementia and traumatic dementia, the cause of AD was unknown, which leads to limited clinical therapeutic measures. With a difficult nursing task and relatively backward domestic nursing experience and concepts, AD nursing remains to be challenging for the nursing doctors, nursing workers, and family members. Active and strategic AD nursing can maximize the function and independence of patients. However, since AD patients were usually nursed at home or nursing home, it was difficult for doctors and nurses to popularize the advanced nursing concepts to AD patients and their nursing staffs. The extended nursing refers to the continuous follow-up and guidance from the hospital to family or community^[3]. In this study, an extended nursing scheme was designed and implemented for patients with moderate and severe AD and their nursing staff. By comparing the behavioral and psychological symptoms and the degree of cognitive dysfunction between the patients receiving the extended nursing and the patients receiving routine nursing, the positive effect and prognostic value of the extended nursing on the AD patients were explored,

which could provide a clinical basis and reference for the extension of the extended nursing in AD, and also open up ideas for nursing programs for other neurology diseases. A total of 102 patients with moderate and severe AD admitted to the People's Hospital of Nanjing Medical University from February 2016 to March 2018 were included in this study. Inclusion criteria were, those patients who met all diagnostic criteria related to moderate and severe AD^[4]; patients with different degrees of behavioral and psychological symptoms; patients whose families voluntarily participated in the study and signed an informed consent. Exclusion criteria were, those patients who had used or stopped antipsychotic drugs 2 w before treatment; patients with serious diseases of heart, liver, lung, kidney, and other important organs, AD patients combined with other mental diseases. According to the random number method, patients were divided into the control group and observation group, 51 cases each. In the control group, there were 23 males and 28 females (aged 63-82 y, with a mean age of 71.43 ± 5.14 y), the course of disease was from 4 to 12 y (mean duration of 7.44 ± 2.15 y). In the observation group, there were 24 males and 27 females (aged 62-84 y, with a mean age of 72.38 ± 4.27 y); the course of disease was from 5 to 10 y (mean duration of 7.13 ± 1.64 y). There was no statistical difference in the baseline data between these two groups. This study was approved by the ethics committee of the hospital.

At discharge, the patients and their families were given routine health education and nursing guidance by the

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nurses according to the condition of the patients. After discharge, they were followed up by telephone one every 3 mo. The follow-up content mainly included self-care ability, mental health and the quality of daily sleep and diet. An extended nursing team was set up, including 1 neurologist and 5 nurses with rich experience in AD nursing. The team members participated in the study actively and were trained with the extended nursing scheme. In addition to the routine nursing measures of the control group, the following extended nursing measures were taken. Implementation of discharge guidance plan, establishment of Wechat group for the discharged patients, opening of hotline, online provision of AD rehabilitation, nursing and medication guidance. The Wechat group and hotline mainly have the following functions, inquiry function. After the patients release or call to raise nursing and medication issues within the group, the designated nursing team members should reply within 1 d and organize appropriate case extension analysis and discussion within the Wechat group. For privacy problems of the individual patients and nursing staff, Wechat private chat or telephone contact was used. When the problems could not be solved remotely, the patients were required to return to the hospital for further consultation. The release of the educational program about AD nursing knowledge and music and video that can assist patients in the treatment. The design of knowledge education plan was considered comprehensively from the aspects of daily life ability training, body function training, language training, memory training. Many training and teaching videos, which were simple and easy to understand, highly operational, were released once a day in the morning and afternoon, respectively. Moreover, the videos were released circularly to enhance the training effect. The nursing staffs were encouraged to release the training results of the patients to the Wechat group for display, and the nursing team used voice or video to give positive encouragement and guidance. Enhancement of communication among doctors, nurses and patients and if the patients or nursing staff do not appear in group chat for a month, the nursing team members inquired the reasons, informed the significance of active participation in group chat and recommended participation. Psychological nursing work included during the follow-up, nursing team members discussed with patients about their history and expectations, listened to the memories, achievements, life expectation and life experience of the patients and their family^[5] to help patients to retrieve their memory and restore hope

for the future. In addition, they comforted and dredged the troubles and pains of their family and the nursing staff and to guide the nursing staff to take countermeasures and to assist patients or nursing staff, who have serious psychological symptoms in the follow-up to seek medical treatment for professional psychotherapy. The follow-up personnel improved the support and encouragement of family decision-makers to AD patients through the introduction of previous cases and psychological counselling^[6]. Organization of offline activities including group activities such as the sand table games and simple intelligent toys^[7] were held every 2 mo to improve patient's yearning for a better life and physical function exercise. The behavioral pathology assessment scale of Alzheimer's disease (BEHAVE-AD)^[8] was used in this study. The behavioral and psychological symptoms of the patients were scored by the doctors. BEHAVE-AD has 7 dimensions, anxiety, fear, hallucination, delusion, emotional disorder, aggressive behavior, behavioral disorder, circadian rhythm disorder, a total of 25 items. The score of each item was from 0 to 3 points. The higher the score the more severe the patient's mental behavior symptoms. Montreal Cognitive Assessment Scale (MoCA)^[9] and mini mental state examination scale (MMSE) were used to evaluate the cognitive function of the 2 groups. There were 8 dimensions (30 scores) in MoCA scale, including naming, language, memory, orientation, attention, executive ability, abstract thinking and delayed memory. The higher the score is, the higher the cognitive ability of the patients. The Barthel activities of daily living (BADL) scale was used for the evaluation of the daily living ability. There were 4 dimensions in BADL scale, including eating, bathing, dressing, and going to the toilet. The score of each dimension was from 1 to 4 points, and the total score was 16 points, the higher the score, the better the basic living activity ability of the patients. SPSS 22.0 software was used to analyze the data, the measurement data were expressed by mean±standard deviation, and T-test was used for comparison. $p<0.05$ means the difference was statistically significant. There was no significant difference in BEHAVE-AD scores between the 2 groups on admission. After 1 y follow-up, the BEHAVE-AD scores of the observation group were significantly lower than that of the control group ($p<0.05$) as shown in Table 1. There was no significant difference in MoCA scores between the 2 groups on admission. After 1 y follow-up, the MoCA scores of the observation group were significantly lower than that of the control group ($p<0.05$, Table 2). There was no

TABLE 1: BEHAVE-AD SCORE OF THE OBSERVATION GROUP AND CONTROL GROUP

Group	N	Admission score	After 1 year follow-up
Observation group	51	19.32±3.53	7.21±2.88
Control group	51	19.47±3.37	10.37±2.17
t		0.219	6.258
p		0.827	0.000

TABLE 2: MoCA SCORE OF THE OBSERVATION GROUP AND CONTROL GROUP

Group	N	Admission score	After 1 year follow-up
Observation group	51	17.76±3.68	14.64±2.38
Control group	51	17.04±3.52	12.59±2.70
t		1.010	4.068
p		0.315	0.000

significant difference in BADL scores between the 2 groups on admission. After 1 y follow-up, the BADL scores of the observation group were significantly higher than that of the control group ($p < 0.05$) as shown in Table 3. AD was a progressive neurodegenerative disease. Due to the gradual loss of the orientation, cognitive function, memory ability, and self-care ability, AD patients suffer from memory deterioration, maze and dementia. Some AD patients may even have the delusion of being killed, violence and other special circumstances and need special care after leaving the hospital. In addition, there were no therapeutic methods that can cure AD totally or improve the symptoms of AD effectively. Drug such as olanzapine, which can improve the memory and control mental symptoms, can only delay the disease^[10]. Drugs such as memantine and donepezil can also enhance the memory and cognitive function by increasing the levels of neurotransmitter acetylcholine. AD patients have a long course of disease, which can be as long as 15 to 20 y. Patients need long-term care for there was a large individual difference in the disease duration^[11]. Nursing AD patients has caused a heavy burden on society and family. The outline of Healthy China 2030 plan focuses on the long-term care for the elderly with dementia at the end of their lives, improving the quality of life of the elderly with dementia, and safeguarding the dignity of the elderly with dementia. However, due to the fact that the AD patients were generally cared by their families or communities, the lack of appropriate care means and experience will aggravate the deterioration of AD patients. How to extend the hospital nursing experience to the families and communities was a subject worthy of clinical consideration. Extended nursing, which concept was born in 1947^[12], refers to

the continuous follow-up and guidance from the hospital to family or community, which guarantees the continuity of the nursing work. The development of the extended nursing in China was relatively slow, and it was still in a small-scale exploratory stage. The outline of China's nursing development plan (2016-2020) issued in 2016 clearly proposed that the extended nursing should be taken as the key development task in the 13th 5 y plan period. And the medical institutions should be encouraged to gradually improve the content and mode of nursing services, so as to provide multiple extended nursing services for the discharged patients. A systematic and operable extended nursing scheme was designed in this study, whose purpose was to extend the nursing intervention mode in real time and permeability. It mainly relied on the Wechat, family visit and telephone to establish the doctor-patient family communication channel, nurse-patient family interaction mode, and patient-patient mutual aid mode to promote the prognosis of AD, so as to make the AD clinical nursing spread from hospital to society and allocate the clinical resources and social resources reasonably. The results showed that the behavioral and psychological symptoms of the patients receiving the extended nursing were significantly reduced compared with the control group and the incidence of anxiety, fear, hallucination, emotional disorder, delusion, aggressive behavior, behavioral disorder and circadian rhythm disorder was reduced and slowed. This may be related to the activities of organizing the patients to play sand table games and simple intellectual toys, discussing with the patients about their history and expectations during the follow-up, listening to the patient's and their family memories, achievements, life expectations and life experiences. Moreover, the results of the MoCA scores showed that the cognitive functions of the patients receiving the extended nursing were significantly better than that of the control group after 1 y of the extended nursing. This may be related to the practices of the music and video in Wechat daily push that can assist the patients in the treatment. It is reported that the music therapy can affect the hippocampus, corpus callosum, nucleus accumbens, auditory cortex

TABLE 3: BADL SCORE OF THE OBSERVATION GROUP AND CONTROL GROUP

Group	N	Admission score	After 1 year follow-up
Observation group	51	14.53±1.77	11.41±1.08
Control group	51	15.02±1.82	9.31±1.52
t		1.378	8.043
p		0.171	0.000

and many other brain regions, and AD was related to the failure and degradation of these brain regions. Therefore, various forms of music therapy can stimulate these brain regions effectively, and slow down their degradation rate, thus delaying the disease^[13]. Finally, the BADL scale was also used in this study to compare the ability of daily life of the patients, and it was found that the ability of daily life of the patients receiving the extended nursing was significantly better than that of the control group in dressing, eating, going toilet and bathing. The extended nursing scheme carried out the daily life ability training, gave encouragement and guidance to AD patients, and responded to the questions of the patients and their families positively, all of which were analyzed and popularized as a case. Received high praise among all the patients and their families. There were still some problems to be solved for the extension of extended nursing in AD, such as the improvement of the supporting system and the establishment of the evaluation system. The extended nursing in this study was provided to patients free of charge. In the process of this study, some problems may occur, such as the fatigue and safety of the nursing staffs, the enthusiasm and satisfaction of the nursing staffs, which may interfere with the results of this study. Therefore, the charging and allocation of nursing staffs need to be improved for the extension of extended nursing in AD.

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Conflict of interests:

The authors declared no conflict of interest.

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Effectiveness of Home Based Cognitive Training Programme on Quality of Life and Caregiver Burden among Elderly Persons with Mild Cognitive Impairment in Selected Community Settings of Puducherry

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Abstract

Ageing is a complex process that decreases the ability of different body systems. It effects both physical and psychological functions and decreases the quality of life. When elderly people were weak physically, this will increase the caregiver burden. Home based cognitive therapy was found to be effective in the management of cognitive functions in the elderly patients. The present study was conducted at selected community settings of Puducherry. A total of 314 elderly participants were recruited for the study after obtaining the written informed consent. After recruiting, the participants were randomly grouped into two groups that is control and intervention groups with 157 participants in each group. Home based cognitive training program was administered to the intervention group individuals. The results of the study supports that the home based cognitive training programme is beneficial in the management of elderly population with mild cognitive impairment. There was significant improvement in the quality of life of the elderly patients and also significant decrease in the care givers burden followed by the intervention. The study recommends further detailed studies in this area to adopt the home based cognitive training programme along with routine medical treatment.

Keywords: Cognition, Home based cognitive training program, Quality of life, Caregivers.

Introduction

Ageing is a complex process that decreases the ability of different body systems. It effects both physical and psychological functions and decreases the quality of life. When elderly people were weak physically, this will increase the caregiver burden.¹ Increase in the age increases the probability of Decline in the

cognition functions deteriorates the general health and decreases the quality of life. Home based cognitive therapy was found to be effective in the management of cognitive functions in the elderly patients.² In the patients with cognitive impairment, the major cognitive functions affected are attention, spatial memory and verbal memory.³ These cognitive domains play a very important role in day today life of an individual. Home based cognitive therapy is simple and effective therapy that helps to restore majority of the cognitive functions.⁴ It is easy to apply and patient friendly as he does not need to stay in the hospital. It decreases the financial burden on the patient and his care givers. Hence, the stress experienced by the care givers of the elderly person will be relieved.⁴ It also reduces other burdens like time

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and effort of the care givers. Earlier studies reported multiple benefits followed by the intervention. Though drug therapies are available, there are not able to restore the cognitive functions effectively and also associated with side effects.⁵ Hence, there is a need for alternative therapy like home based cognitive therapy. The present study was undertaken to observe the efficacy of the home based cognitive training programme on quality of life and caregiver burden among elderly persons with mild cognitive impairment in selected community settings of Puducherry.

Materials and Method

Study Design: Experimental study

Study Setting: The present study was conducted at selected community settings of Puducherry.

Study Participants: A total of 314 elderly participants were recruited for the study after obtaining the written informed consent. Convenient sampling technique was used in the selection of the participants. The following criteria were used while recruiting the participants.

Inclusion Criteria:

1. Client with Age ≥ 60
2. Client whose score in MMSE greater than 20
3. Clients who are a permanent resident of the locality for at least in the past 6 months.
4. Clients who are willing to participate in the study

Exclusion Criteria:

1. Clients with significant neurological deficit
2. Client who are in antidepressants
3. Clients with history of alcohol abuse
4. Clients with significant physical disability

After recruiting, the participants were randomly grouped into two groups that is control and intervention groups with 157 participants in each group.

Home Based cognitive Training Programme:

Each training session comprised the following steps

1. **Orientation in time and space:** Performed using external aids such as a calendar and the day's newspaper, whereby participants determined the current day, month and year.

2. **Presentation of the Names:** Participants and the researcher presented their names. Verbal associations were elicited between the person's name and their appearance.

3. **Visual and Auditory Attention Exercises:** The visual attention task entailed identifying details in photographs, letters or figures amongst several stimuli spread out on a single printed sheet, as well as spotting differences between two similar photographs, among other activities. The auditory attention exercises included tasks such as detecting words in songs and identifying whether two consecutively repeated sequences of numbers or words matched or differed.

4. **Memory Exercises Using Visual Aids:** The categorization strategy with ecological tasks was used, i.e. simulated activities of daily living, graded from simple to more complex. Grocery items were used in the early stages of training progressing to supermarket item lists only, by the end of training.

5. **Transfer Task:** Practical tasks from activities of daily living were used, such as to the supermarket, giving and checking change, etc. The older adults were expected to calculate the total cost of the purchase and offer change or check the change received.

Cognitive training was given to all eligible subjects individually two sessions weekly continuously for a period of 1 month i.e. a total of 8 sessions. Each session will take approximately 1 hour.

Ethical Considerations: The present study protocol was approved by the institutional human ethical committee. The study was conducted as per the guidelines of ICMR.

Data Analysis: Data was analysed using SPSS 20.0. Data was expressed in frequency and percentage and mean and SE. Paired and unpaired t test was used to observe the significance of difference. Probability value less than 0.05 was considered as significant.

Results

Results were presented in table no 1. There was a significant decrease in the care giver burden in the experimental group when compared with the control group. There was a significant improvement in the quality of life in the individuals of experimental group when compared with control group.

Table 1: Care giver burden and quality of life of control and experimental groups.

S.No.	Parameter	Group	Mean±SE	Paired 't'-test		Un paired 't'-test	
				Con- pre test -1st post test	Experimental pre -1st post	Con- pre test- experimental pre test	Control post – experimental post
1.	Care giver burden	Control pre test	55.36±0.45	t=2.22 p=0.03*		t=1.98 p=0.048*	t=60.11 p<0.001***
		Control 1st post test	53.87±0.49				
		Experimental pre test	54.14±0.43		t=65.55 p<0.001***		
		Experimental 1stpost test	19.61±0.29				
2.	Overall Quality of life	Control pre test	180.88±1.67	t=0.678 p=0.498		t=0.272 p=0.785	t=42.73 p<0.001***
		Control 1st post test	182.36±1.82				
		Experimental pre test	180.24±1.64		t=48.03 p<0.001***		
		Experimental 1stpost test	279.73±1.37				

*P<0.05 is significant; **P<0.01 is significant: ***P<0.001 is significant

Discussion

The present study was undertaken to observe the efficacy of the home based cognitive training programme on quality of life and caregiver burden among elderly persons with mild cognitive impairment in selected community settings of Puducherry. The results of the study supports that the home based cognitive training programme is beneficial in the management of elderly population with mild cognitive impairment. There was significant improvement in the quality of life of the elderly patients and also significant decrease in the care givers burden followed by the intervention. The study recommends further detailed studies in this area to adopt the home based cognitive training programme along with routine medical treatment. It was reported that about 15 million people are providing care giving to the elderly population.^{6,7} Interestingly, majority of these care givers are family members who are sparing their time and losing their earnings and undergoing immense stress. Further, hospitalization is most common in the elderly patients with impairment of cognition.⁸ During hospitalization, physical, psychological, social and financial aspects of burden is involved.⁹ Impairment of cognition in elderly persons is most common and further if untreated, it can be converted as severe diseases like Alzheimer's disease.¹⁰ The available drugs and other medical treatments are costly and difficult to afford them by the average individual. Hence, the need for a non-pharmacological therapies are essential for the

management of the impairment of cognitive functions.¹¹ Home based cognitive training program was considered as essential and better therapy in the management of the cognitive decline as it does not have contra indications and affordable.¹² Earlier studies reported that home based cognitive training program improves the quality of life of the participants.¹³ It also increases the memory and cognition of the patients so that their dependency on the care givers will be decreased. This will reduce the burden on the care givers.^{14,15} Interestingly, home based cognitive training program was found to be very effective in improving the cognition in both the healthy and patients with cognitive decline.¹⁶⁻¹⁸ The present study agrees with earlier studies as we have observed improvement in the quality of life and decline in the care givers burden.

Conclusion

The results of the study supports that the home based cognitive training programme is beneficial in the management of elderly population with mild cognitive impairment. There was significant improvement in the quality of life of the elderly patients and also significant decrease in the care givers burden followed by the intervention. The study recommends further detailed studies in this area to adopt the home based cognitive training programme along with routine medical treatment.

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Effectiveness of Resilience Education in the Mental Health of Family Caregivers of Elderly Patients with Alzheimer's Disease

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Abstract

Background: Continuing care of an elderly patient with Alzheimer's disease may result in psychosocial and physical disorders in family caregivers, as well as limitations to providing complete care.

Objectives: The aim of the present study was to determine the effectiveness of resilience education in the mental health of family caregivers of elderly patients with Alzheimer's disease.

Methods: This study followed a randomized, clinical trial design. The study population consisted of family caregivers of elderly patients with Alzheimer's disease who referred to hospitals and neurologists' offices located in the western cities of Mazandaran province, Iran. Data were collected between 2016 and 2017. Fifty-four family caregivers were recruited through convenience sampling and randomized into control and intervention groups (27 in each group). Demographic characteristics, Connor-Davidson Resilience Scale, and General Health Questionnaire were used for data collection. Resilience education was provided in eight sessions (45 min each) using PowerPoint presentations and educational pamphlets. The mental health scores were calculated using the indices including normality tests and analyzed using the independent and paired *t*-tests, analysis of variance (ANOVA), and Bonferroni test.

Results: An independent *t*-test showed improvements in the mental health status of those who received the educational intervention (mean difference: 23.8 ± 6.4 ; $P < 0.001$). The independent *t*-test also indicated statistically significant differences between the groups' anxiety/insomnia (8.8 ± 2.3 versus -0.12 ± 2.5), somatic symptoms (7.2 ± 2.2 versus -0.9 ± 1.03), social dysfunction (4.5 ± 2.9 versus 0.2 ± 1.3), and depression (3.2 ± 2.2 versus -0.08 ± 0.6 ; $P < 0.001$).

Conclusions: Resilience education successfully improved the mental health of family caregivers. Therefore, it is suggested that healthcare providers, Alzheimer's associations, and NGOs provide educational interventions to help promote the caregivers' mental health.

Keywords: Alzheimer's Disease, Resilience, Psychological, Family Caregivers

1. Background

Alzheimer's disease is a common chronic disease among elderly patients (1, 2), which results in progressive memory loss, personality change, and difficulty in daily activities. It is estimated that approximately 100 million elderly patients will live with Alzheimer's disease by 2050 (3). There are no official statistics regarding the disease in Iran; however, according to the vice-president of the Iranian Alzheimer's Community, there are currently 212000 patients with Alzheimer's disease in Iran who require at least one family caregiver (4). Research shows that family caregivers meet more than 80% of Alzheimer's patient needs. Often, the elderly person can be cared at home

by favorable levels of caring services. Therefore, isolation, physical fatigue, and mental pressure threaten family caregivers (2-4). Studies have revealed that more than 80% of family caregivers experience high levels of social, physical, and psychological problems, such as stress, depression, and anxiety, alongside coping with the caring pressure (3, 4). Therefore, the majority of family caregivers require knowledge and skills to promote self-care and health.

Numerous solutions are recommended for promoting family caregivers' health. Building resilience is suggested and describes a situation in which a caregiver improves social performance and overcome difficulties, despite experiencing high mental pressure (3). Santos et al. showed that resilience education could decrease stress and promoted

caregivers' physical and psychological health (1). Other research has found that the employment of resilience training offers many advantages (5-7). Many resilience studies explore personal characteristics that may affect coping or resilience and focus on individuals experiencing specific adverse circumstances (e.g., illness, bereavement, abuse, etc.). Researchers have demonstrated the effectiveness of interventions that assist family caregivers to provide support for patients with dementia (8-10).

2. Objectives

While there are many studies examining stress management as part of diminishing the pressure of caregiving (2, 6, 8-10), we were not aware of any studies that used resilience training to promote mental health in family caregivers of elderly Alzheimer's patients in Iran. Therefore, this study aimed to determine its efficacy in this population.

3. Materials and Methods

This study followed a randomized, clinical trial design. The study population consisted of family caregivers of elderly Alzheimer's patients who referred to hospitals and neurologists' offices located in the western cities of Mazandaran province, Iran (Ghaemshahr Razi Hospital; Ramsar, Tonekabon, and Ghaemshahr neurologists' offices). Data were collected between 2016 and 2017. Having considered a 95% confidence level, a power of 0.8 for a two-tailed test, and expecting a 10 percent difference in the mental health mean score based on similar studies (2, 6), the required sample size was calculated as 64 participants (32 people for each group). Participants were recruited through convenience sampling. In total, 64 family caregivers of older adults with Alzheimer's disease completed an initial eligibility assessment. However, 10 family caregivers were excluded from the study based on the exclusion criteria ($n = 4$) or due to declining to participate ($n = 6$). The remaining 54 eligible participants were grouped into intervention and control groups by a block-randomized allocation method. The participants in odd-numbered blocks were placed into the intervention group ($n = 27$) and even-numbered blocks were placed in the control group (i.e., no intervention group; $n = 27$). The caregivers in the groups were adjusted for age, sex, marital status, occupation, education level, income level, caring duration, smoking situation, and family relationship with the patient. Two participants in the intervention group missed two or more program sessions and two participants in the control group missed assessments due to the death of their patients during the program. Therefore, of 54 family caregivers, 50

(92.6%) completed all follow-up assessments (25 in the intervention group and 25 in the control group).

Family caregivers of elderly patients with Alzheimer's disease were invited if they were not caring for any other patients, were giving full-time and complete care in three domains of financial care (paying for treatment and caring), physical care (helping in patient's daily activities), and emotional care (controlling patient's fear, anxiety, and delirium) (11), were able to respond to questions, and were literate. The intervention could be discontinued if the caregivers missed at least two intervention sessions, the patient died, or the participant revoked their cooperation. The data collection tools included:

1. The demographic characteristics questionnaire (including age, sex, marital status, occupation, education level, income level, caring duration, smoking habits, and family relationship with the patient).

2. Connor-Davidson Resilience Scale: This instrument is based on a five-point Likert scale (from 0 = absolutely correct to 4 = always correct). The reliability and validity of this questionnaire were approved in Besharat's research (11) with a Cronbach's alpha of 0.84. The questionnaire was sent to 10 faculty members for validity assessment and administered to 30 caregivers for reliability assessment. Internal consistency was approved by Cronbach's alpha of 0.87 and stability with test-retest method ($r = 0.92$). The demographic characteristics and resilience questionnaires were completed before the intervention in both groups.

3. General Health Questionnaire (GHQ): The GHQ consists of 28 items and four subscales (somatic symptoms, anxiety/insomnia, social dysfunction, and severe depression). Participants receiving a cutoff score of 23 or higher were considered to have probable psychiatric distress (11, 12). The GHQ was completed before and 35 - 40 days after the intervention in both groups.

The intervention was administered as a training program that aimed to promote resilience development through PowerPoint presentations and educational pamphlets. These were supplied through a lecture for the intervention group. The program was based on Henderson-Milstein and Kravets' resilience education projects, as presented in Table 1 (11, 12). The group-based intervention process required eight sessions (45-minute sessions once per week) and was held as an educational class by one of the researchers in a hospital located in Mazandaran province, Iran. There was no intervention designed for the control group and the participants in this group received routine care, such as education on medicinal and non-medicinal care for their patients. At the end of the intervention, an educational program was implemented that aimed to promote resilience development in the control group.

Participants' written consent was obtained and all data were kept anonymous and confidential. Participants

Table 1. Outline of Training Program

Session	Educational Content
1	Introducing the members; Explaining Alzheimer's disease, resilience and individuals specifications
2	In-person supportive factors (self-esteem, optimism, etc.)
3	External supportive factors and personal sense of responsibility
4	Resilience promotion solutions (commitment)
5	Resilience promotion solutions (control)
6	Resilience promotion solutions (challenging)
7	Resilience promotion solutions (coping)
8	Concluding the discussed topics (problem solving)

were assured that their participation was voluntary and that they could withdraw from the study at any time without any negative repercussions. The researchers ensured that ethical standards were adhered to during and after data collection. The study protocol was approved by the Research Ethics Committee of Babol University of Medical Sciences (reference number: MUBABOL.REC.1395.48). Data were analyzed using SPSS version 16. The normality of data was evaluated using the Kolmogorov-Smirnov test. The mental health score means were analyzed using parametric tests, such as independent and paired *t*-tests, analysis of variance (ANOVA), and Bonferroni test.

4. Results

The average age was 43.4 ± 6.3 years in the control group and 42.6 ± 6.2 years in the intervention group. The study sample demographics are shown in Table 2. The Shapiro-Wilk test was used to check normality before and after intervention in both groups and the data were considered to be normal at a significance level of 0.05. The mental health variable in both groups showed a normal distribution before ($P = 0.94$) and after the intervention ($P = 0.53$). The variance homogeneity was checked using Levene's test ($P > 0.05$). The variances were homogeneous at pretest while they were heterogeneous at posttest. However, the equality of variances hypothesis was accepted since the overall standard deviation was smaller than the mean.

The results showed that there was a significant difference ($P < 0.001$) between the mental health scores before (39.9 ± 8.4) and after (16.08 ± 3.7) the intervention in the intervention group. A paired *t*-test was used to compare the scores of mental health subscales. There was a significant difference in somatic symptoms, anxiety/insomnia, social dysfunction, and severe depression scores before and after the intervention ($P < 0.001$; Table 3).

An ANOVA test was used to demonstrate the hypothesis that "the mental health condition of family caregivers varies with their demographic characteristics". The effects

Table 2. Demographic Variables of Intervention and Control Groups Before Intervention

Variables	Control	Intervention	P Value ^a
Marital status			0.57
Single	4 (16)	5 (20)	
Married	14 (56)	18 (72)	
Widow/widower	5 (20)	1 (4)	
Divorced	2 (8)	1 (4)	
Gender			0.22
Female	18 (70)	22 (88)	
Male	7 (30)	3 (12)	
Education level			0.23
Primary education	9 (36)	3 (12)	
High school diploma	11 (44)	8 (32)	
University education	5 (20)	14 (56)	
Employment status			0.16
Retired	3 (12)	6 (24)	
Self-employed	8 (32)	5 (20)	
Employee	3 (12)	7 (28)	
Worker	2 (8)	0 (0)	
Housewife	9 (36)	7 (28)	
Income level			0.49
Low	4 (16)	0 (0)	
Moderate	16 (64)	14 (56)	
High	5 (20)	11 (44)	
Smoker			0.31
Yes	4 (16)	7 (28)	
No	21 (84)	18 (72)	
Caring duration			0.48
6 months	0 (0)	2 (8)	
6 - 24 months	12 (48)	15 (60)	
> 24 months	13 (52)	8 (32)	
Family relationship			0.45
Child	20 (80)	22 (88)	
Spouse	5 (20)	3 (12)	

^aThe significance level was 0.05.

of age ($F = 5.85$, $P = 0.003$), income level ($F = 3.3$, $P = 0.04$), and caring duration ($F = 3.5$, $P = 0.03$) were significant on caregivers' mental health while the variables of marital status, education level, smoking habits, and family relationship with the patient had no significant effect ($P > 0.05$). Pairwise comparisons using the Bonferroni test showed that an increase in age, income, and caring duration decreased mental health scores, indicating that aging resulted in improved mental health. The mental health of caregivers with lower incomes was significantly different from the mental health of those with higher incomes, as a higher income resulted in a lower mental health score. The

Table 3. Comparison of the Mean Scores of Mental Health and Related Subscales in the Two Groups^a

Mental Health Subscales, Groups	Pre-Intervention	Post-Intervention	P Value ^b	Mean Difference	P Value ^b
Severe depression					< 0.001
Intervention	4.5 ± 3.04	0.4 ± 0.7	< 0.001	3.2 ± 2.2	
Control	4.8 ± 3.6	5.08 ± 3.7	0.23	-0.08 ± 0.6	
Anxiety/insomnia					< 0.001
Intervention	11.6 ± 2.8	3.4 ± 1.5	< 0.001	8.8 ± 2.3	
Control	10.04 ± 4.02	10.5 ± 3.7	0.16	-0.12 ± 2.5	
Somatic symptoms					< 0.001
Intervention	10.8 ± 3.2	3.5 ± 1.5	< 0.001	7.2 ± 2.2	
Control	8.9 ± 2.7	9.8 ± 2.7	0.088	-0.9 ± 1.03	
Social dysfunction					< 0.001
Intervention	12.9 ± 4.3	8.6 ± 2.2	< 0.001	4.5 ± 2.9	
Control	8.2 ± 3.4	7.4 ± 3.01	0.12	0.2 ± 1.3	
Mental health					< 0.001
Intervention	39.9 ± 8.4	16.08 ± 3.7	< 0.001	23.8 ± 6.4	
Control	32.04 ± 7.02	32.9 ± 7.2	0.11	-0.9 ± 2.8	

^aValues are expressed as mean ± SD.^bThe mean difference is significant at the 0.05 level.

results also showed that the mental health of those caring for their family members for six months was significantly better than the mental health of those who cared for their family members for 24 months or more. In this regard, the logistic regression analysis showed that the odds ratio of caring duration was 6.76 ($P = 0.04$). This indicated that caregivers with longer caring duration could experience mental health difficulties 6.76 times those with shorter caring durations.

5. Discussion

The results indicated that resilience education promoted the mental health of family caregivers of elderly patients with Alzheimer's disease, which is similar to other studies (2, 13, 14). This could be explained by stating that resilience education for caregivers was successful in inducing the feeling of strength for dealing with high levels of stress. Martin-Carrasco et al., also believed that educational interventions resulted in more effective interactions in caring among caregivers and that teaching them on how to find better solutions in facing adversity decreased their physical and mental problems (13).

The resilience education was also successful in decreasing anxiety/insomnia and severe depression, which are also consistent with other studies (2, 14, 15). The results by Hosseini Ghomi's confirmed the effectiveness of resilience training in stress reduction of mothers whose children suffered from cancer in Imam Khomeini Hospital of Tehran (16).

In fact, resilience educational programs have been found to be effective in improving coping mechanisms (2).

While our findings are consistent with those in the literature showing that higher levels of resilience were associated with lower depression rates and greater physical health (2, 14, 15), they are in contrast to Moljord's results that showed a negative correlation between resilience, physical activity, and depressive symptoms in adolescents (17).

The results showed that resilience education was successful in decreasing somatic symptoms and social dysfunction among family caregivers. Dias' study also showed that less resilient caregivers suffered from physical problems and referred to doctors more often than did those with higher resilience. Resilience increased self-care among caregivers, positively changed their habits and lifestyle, made them stronger in facing stressors of caregiving, and decreased the probability of experiencing physical symptoms (2). Additionally, more resilient caregivers were reported to have better social interactions and more success in social activities (2, 13, 18).

Fernandez-Calvo et al.'s study showed interventions that encouraged active coping techniques had the best effects on caregivers in terms of reducing the impact of adversities generated during care, creating self-confidence to move forward, on augmentation their experience of positive emotions, and promoted self-efficacy and competence in providing care (19). Losada et al.'s study showed that commitment therapy for dementia family caregivers minimized negative or dysfunctional thoughts resulting from the care (20).

Our results also showed a significant relationship between the caregiver's age and mental health. Aging was found to increase resistance to stress load and helped determine more effective solutions in facing adversity. Adults are typically excluded from policymakers and healthcare providers' health promotion programs. Therefore, family caregivers who are often the patients' children and are at the middle age or older may be neglected, while different, multi-aspect health-threatening factors can influence them. Fitzpatrick and Vacha-Haase's study showed an inverse relationship between age and mental health and older caregivers had higher life satisfaction and were experiencing less psychological problems (15).

Our results showed a significant relationship between caring duration and caregiver mental health. Caregivers could be threatened with social activity dysfunction, fatigue, or exhaustion as the caring duration increased. Therefore, providing social-based services (e.g., respite services, daily care, transportation, and emergency responding services) are suggested as a solution for mental health promotion. Other research has also shown that previous experience or long-lasting elderly care, the family's social and financial status, knowledge and information about the disease, religion, and governmental support are effective in improving caregivers' mental health (3).

Based on the present study, we concluded that a significant relationship existed between income level and mental health, which is consistent with observations in previous studies. For instance, Dias et al. and Martin-Carasco et al. suggested that higher income was an indicator of better-coping mechanisms when facing problems and decreased psychological disorders among caregivers (2, 13).

Based on our results, mental health promotion after the intervention is not affected by the caregivers' sex. This could be explained by changes in Iranian culture; sons now have more roles in elderly caring than before. The equal sex distribution among our study groups might be another explanation. This is in contrast to the results of Clay that suggested sex and family relationship as variables affecting the caring pressure among caregivers of elderly patients with Alzheimer's disease. He also claimed that female caregivers experienced less depression, stress, and anxiety than males (21).

There was no significant correlation between family relationship and mental health in our study. The reason might be that the majority of the caregivers participating in this study were the children of their patients. This finding is in contrast to the results of Kang, who showed that spousal caregivers were more resilient than children or other relatives (18). de Oliveira Gaioli's study also discussed the role of family relationship and caring duration in caregivers' mental health (21). These differences can be explained by discrepancies in the research popu-

lation and setting. Since the spouses of elderly patients with Alzheimer's disease often suffer from a chronic disease, this limits them in providing care and results in the transfer of caring pressure to their children.

The smoking behavior was another ineffective factor in caregivers' mental health, as the majority of our participants were nonsmokers. Andren and Elmstahl's results showed that unsafe behaviors, such as smoking, could induce physical and mental disorders; therefore, smoking caregivers were less resilient than were nonsmoking caregivers (22).

The education level was not significantly related to the mental health of caregivers of Alzheimer's disease patients, as our results showed. This can be due to that mental health promotion is dependent on health literacy rather than education level. However, this finding is in contrast to the results of Leach et al. (6). Marital status was another irrelevant factor found in this study, while Andren and Elmstahl considered married caregivers to have more social support networks, resulting in less stress and better mental health (20).

5.1. Limitations

Our small sample size may have resulted in a study sample that is not representative of the general population. Moreover, convenience sampling was used in the present study. Since the caregivers were easily available, this sample was not representative of the general population of elderly Alzheimer's disease patients' family caregivers. Future research should employ random sampling to increase generalizability.

5.2. Conclusions

In the present study, resilience education successfully promoted the mental health of family caregivers. Therefore, educational interventions provided by healthcare providers, the Alzheimer's associations, and NGOs can promote caregiver mental health. Financial support and shortening the caring duration by including other family members are among the solutions recommended for caregivers' mental health promotion. Healthcare providers, especially nurses, are more suitable to provide society-based services and practical solutions for the mental health promotion in this population.

This study is among the limited studies attempting to determine the role of coping strategies in promoting mental health in caregivers of elderly Alzheimer's disease patients. Therefore, we suggest that interventional studies be conducted using coping strategies against psychological disorders and mental pressure/stress in this group of clients. There was no significant relationship between mental health and family relationship with the patient, education level, smoking habit, and occupation; thus, future

studies investigating these factors are needed for more clarification.

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Footnotes

Authors' Contribution: Fatemeh Ghaffari, Zahra Fotokian, and Mehdi Rostami studied the concept and designed it. Fatemeh Ghaffari and Zahra Fotokian analysed and interpreted the data. Zahra Fotokian and Fatemeh Ghaffari drafted the manuscript and revised the manuscript critically for important intellectual content. Mehdi Rostami and Mahmoud Hajiahmadi analyzed the data statistically.

Clinical Trial Registration Code: The current study was registered in the Iran Registry of Clinical Trials (IRCT2016080829251N1).

Conflict of Interests: The authors declare no conflict of interest.

Ethical Approval: The study protocol was approved by the University of Babol Medical Sciences research ethics committee (Reference number: MUBABOL.REC.1395.48).

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Clinical Trial Protocol

Iranian Registry of Clinical Trials

08 Dec 2022

The effect of resilience training on the mental health of family caregiver of elderly with alzheimer

Protocol summary

Summary

Aim: The effect of resilience training on mental health of family caregiver of elderly with Alzheimer. The study population: family caregivers of elderly with Alzheimer who live in the Ghaemshahr city. Study design: randomized clinical trial. Sample size: totally 50 patients (25 patients in each group) were participated. The individuals were selected based on the inclusion and exclusion criteria and randomly assigned into two intervention and control groups. Mental health questionnaire was filled out for patients by researcher in both groups, before and after intervention. In this study, resilience training was delivered to the participants through 45-minute educational sessions and pamphlet. These sessions were held once a week for 8 weeks in Razi hospital. The control group received no intervention. After the training program (40 to 45 days later) mental health questionnaire was filled out again for participants in both groups.

General information

Acronym

IRCT registration information

IRCT registration number: **IRCT2016080829251N1**

Registration date: **2017-02-11, 1395/11/23**

Registration timing: **retrospective**

Last update:

Update count: **0**

Registration date

2017-02-11, 1395/11/23

Registrant information

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Recruitment status

Recruitment complete

Funding source

Vice chancellor for research, Babol University of Medical Sciences

Expected recruitment start date

2016-02-09, 1394/11/20

Expected recruitment end date

2016-08-10, 1395/05/20

Actual recruitment start date

empty

Actual recruitment end date

empty

Trial completion date

empty

Scientific title

The effect of resilience training on the mental health of family caregiver of elderly with alzheimer

Public title

The effect of resilience training on mental health

Purpose

Supportive

Inclusion/Exclusion criteria

Inclusion criteria: Resilience score is less than 50, but The other does not take care to elderly with Alzheimer's, Carers aged at least 30 and a maximum of 55 years, Full care in three areas of financial, emotional and physical done Having the ability to answer the questions. Exclusion criteria: unwillingness to cooperate and Not participating in In two sessions of training sessions.

Age

From **30 years** old to **55 years** old

Gender

Both

Phase

N/A

Groups that have been masked

No information

Sample size

Target sample size: **50**

Randomization (investigator's opinion)

Randomized

Randomization description

Blinding (investigator's opinion)

Not blinded

Blinding description

Placebo

Not used

Assignment

Parallel

Other design features

Secondary Ids

empty

Ethics committees

1

Ethics committee

Name of ethics committee

Ethics Committee of Babol University of Medical Sciences

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Babol

Postal code

Approval date

2016-06-05, 1395/03/16

Ethics committee reference number

IR.MUBABOL.REC.1395.48

Health conditions studied

1

Description of health condition studied

Alzheimer

ICD-10 code

G30

ICD-10 code description

Alzheimer disease

Primary outcomes

1

Description

Resilience

Timepoint

Before and 40 days after intervention

Method of measurement

Resilience questionnaire

Secondary outcomes

1

Description

Mental health

Timepoint

Before and 40 days after intervention

Method of measurement

Mental health questionnaire

Intervention groups

1

Description

Intervention group: Resilience training was delivered to the participants through educational sessions and pamphlet. These sessions were held once a week for 8 weeks.

Category

Behavior

2

Description

Control group: individuals in this group didn't receive any intervention.

Category

Treatment - Drugs

Recruitment centers

1

Recruitment center

Name of recruitment center

Razi hospital and doctors neurologists' offices in Ghaemshahr

Full name of responsible person

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Street address

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City

Sari

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1

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Name of organization / entity

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Full name of responsible person

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Grant name
Grant code / Reference number
 9542031
Is the source of funding the same sponsor organization/entity?
 Yes
Title of funding source
 Vice Chancellor for research of Babol University of Medical Sciences
Proportion provided by this source
 100
Public or private sector
 empty
Domestic or foreign origin
 empty
Category of foreign source of funding
 empty
Country of origin
Type of organization providing the funding
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Sharing plan

Deidentified Individual Participant Data Set (IPD)
 empty
Study Protocol
 empty
Statistical Analysis Plan
 empty
Informed Consent Form
 empty
Clinical Study Report
 empty
Analytic Code
 empty
Data Dictionary
 empty

The Impact of Educational Intervention on the Anxiety of Family Caregivers of the Elderly with Dementia: A Randomized Controlled Trial

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ABSTRACT

Background: Due to the increasing population of elderly and the consequent increase in the number of chronic diseases such as dementia, the psychological complications such as anxiety in the family caregivers increase. The aim of this study was to determine the effect of educational intervention on the anxiety of family caregivers of the elderly people with dementia.

Methods: This randomized controlled trial was performed in the elderly Neurology Clinics in Shiraz from May to August 2017. This study was conducted on 70 families with elderly people with dementia who were randomly divided into an intervention (receiving in groups for seven sessions of educational intervention) and a control group (Conventional care). Data collection tool was Spielberger's Anxiety Inventory (40-items, score=20-80). Data were analyzed by SPSS version 21 using ANOVA test with repeated measures and independent t-test.

Results: One and three months after the interventions, the mean scores of anxiety in the intervention group were 70.51 ± 3.78 and 70.31 ± 3.43 and in the control groups they were 76.45 ± 3.45 and 76.22 ± 5.08 respectively. The results showed significant differences between the two groups regarding anxiety after the intervention ($P < 0.001$).

Conclusion: Educational programs held to promote and maintain the physical and mental health of caregivers could reduce the anxiety of family caregivers of the elderly with dementia.

Trial Registration Number: [IRCT2017080915426N4](https://www.irct.ir/record/IRCT2017080915426N4).

KEYWORDS: Anxiety, Dementia, Education, Family caregivers

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INTRODUCTION

According to the World Health Organization (2016), 47.5 million people worldwide suffer from dementia and it will reach 76 million by 2030.^{1,2} Precise statistics on the prevalence of dementia in Iran are not available. It is estimated that over 500,000 people in Iran suffer from Alzheimer's type of dementia.³ Elderly diseases, such as dementia, are expected to increase as age grows.⁴ Dementia is a long-term, progressive syndrome¹ in which the level of consciousness in the individual does not change and causes disruptions in behavior, ability to perform daily activities, thinking and memory.⁵ ⁶ In this case, there is a sharp decrease in the elderly's independence, and an increase in their dependence on functions will be observed.³ The elderly with dementia generally require a high level of long-term care.⁷ These problems will lead to dependence and the urgent need for support by caregivers.⁸ Approximately 85% of people with dementia are cared for by their families at home.⁹ In many cases (57% -81%), women between the ages of 45 and 70 spend 12 hours daily taking care of a patient with dementia. Therefore, family caregivers of the elderly people with dementia are at high risk of losing mental and physical health. More than 80 percent of Alzheimer's caregivers have experienced high levels of anxiety,¹⁰ and about one third (25-30 percent) of dementia caregivers have reported clinical signs of anxiety. Anxiety accompanied by feelings of guilt and fatigue in caregivers that do not have adequate medical information about dementia and its usual symptoms and previous experience in how to care for these patients has been observed. Sometimes, caregivers often become anxious because they may not see the effects of their work or think that their care was not enough.¹¹⁻¹³ These harmful cases affect the well-being of the caregivers and also the ability to fully take care of the patient.⁸

In addition, caregivers of the elderly people with dementia have a negative attitude in terms of physical health, energy, mood, memory and ability to perform

recreational activities.¹⁴ About half of the caregivers are suffering from high blood pressure. Arthralgia, obesity and limitation in activities, the prevalence of diabetes and high blood cholesterol in caregivers of the elderly people with dementia have also been reported. Reduction of the quality of life of caregivers reduces the quality of care for the elderly and has a negative effect on the progress of symptoms of dementia in the patient.^{15, 16}

In order to prevent and reduce the negative side effects in caregivers of the elderly with dementia and to improve their health and quality of life, many studies have been conducted on the problems of caregivers and their lack of support, but the number of studies conducted for reducing anxiety of the elderly caregivers with dementia is limited. In addition, caregivers often have not received any formal education for dementia or for themselves. Since lack of adequate knowledge about illness and caring skills could lead to anxiety and exhaustion in caregivers and lack of enough knowledge of caregivers about self-care could also lead to physical and psychological complications and a decrease in their quality of life.^{8, 16} Therefore, the aim of this study was to determine the impact of on-site educational intervention along with informational support on the anxiety of family caregivers of elderly people with dementia.

MATERIALS AND METHODS

This randomized, controlled trial was conducted from May to August 2017. The study was conducted in geriatric clinic in Imam Reza and other neurology clinics affiliated to Shiraz University of Medical Sciences, Shiraz, Iran. The target population included family caregivers of the elderly people with dementia living in Shiraz. Inclusion criteria for caregivers included being the patient's main caregiver and taking on the main responsibilities of the patient; being over 18 years old; having reading and writing skills; having a three-month experience of patient care; access to telephone, mobile or social networks; and obtaining anxiety score

more than 31 from Spielberger's Anxiety Inventory and lower than 80.

The elderly person should also have a history of at least six months to five years of dementia. Exclusion criteria for the caregivers included getting physical and mental illness during the study which prevents patient care and drug abuse; using alcoholic drinks or psychotropic drugs; occurrence of a critical event or severe stress such as the death of closed ones, divorce; etc. during the study or in the last 3 months for the caregiver, participating of caregiver in training sessions on the care of the elderly people with dementia in the last 6 months; and taking care of someone else in the family. In addition, the death of the elderly person with dementia during the study, the history of the elderly in having other well-known chronic and debilitating diseases such as heart failure, uncontrolled asthma, uncontrolled diabetes, etc., before or during the study and transfer of the elderly patients with dementia to long-term care centers at the time of study were the other conditions of exclusion from the study.

In this study, the sample size was estimated according to the study carried out by Lynn et al.¹⁷ For smaller effect sizes, we needed to enhance the sample size. Therefore, based on the following formula a 62-caregivers sample size was determined. Then, it was raised to 70 (35 subjects in each group).

$$n1 = n2 = \frac{(Z_{1-\alpha/2} + Z_{1-\beta})^2 * s_d^2}{d^2} = 35$$

$$\beta=0.2-S_1=7.16-S_2=7.39-\alpha=0.05-\mu_1=24.75-\mu_2=21.08$$

The study participants were selected through convenience sampling when they referred to the neuropsychology and geriatric specialty clinic. Then, they were allocated to the intervention and control groups using block randomization with a random sequence of 2 block sizes. This was done through block design software. Each block was placed in a sealed package and sampling continued until the end. 85 caregivers were enrolled into the study. All subjects were invited to participate

in the preliminary session to obtain their initial information and then, based on the inclusion criteria, they were evaluated, and eligible caregivers were selected. However, ten caregivers did not have the inclusion criteria, and five patients were not willing to participate in the study. Therefore, the study was performed on 70 subjects divided into intervention and control groups. It should be mentioned that all the 70 subjects finished the study (Figure 1).

The intervention group was provided with educational intervention and a scientifically valid educational book approved by faculty members in Shiraz University of Medical Sciences. The control group did not receive any formal education regarding the care of the elderly with dementia, during the study. In order to have ethical considerations in the research, after conducting the study, the educational programs were carried out and also, a book was provided for them.

The caregivers signed written informed consents. Besides, they received verbal and written information about the study objectives, design, and duration. They were also reassured about their right to withdraw from the study. The confidentiality and anonymity of the caregivers' information were considered, as well.

In this study, the assistant who collected the data and the statistician who analyzed the data were blind to the study groups. It should be noted that the caregivers in the intervention group were asked not to explain the program to other caregivers.

The study was approved by the Ethics Committee of the Shiraz University of Medical Sciences with the code of IR.SUMS.REC.1396.76.

The research included two stages of educational intervention and information stabilizing period. In the first stage, the way of caring an elderly with dementia and caring for her/himself were prepared in seven sessions of one and half hours and delivered by lectures and using educational aids such as PowerPoint, video projector, video player

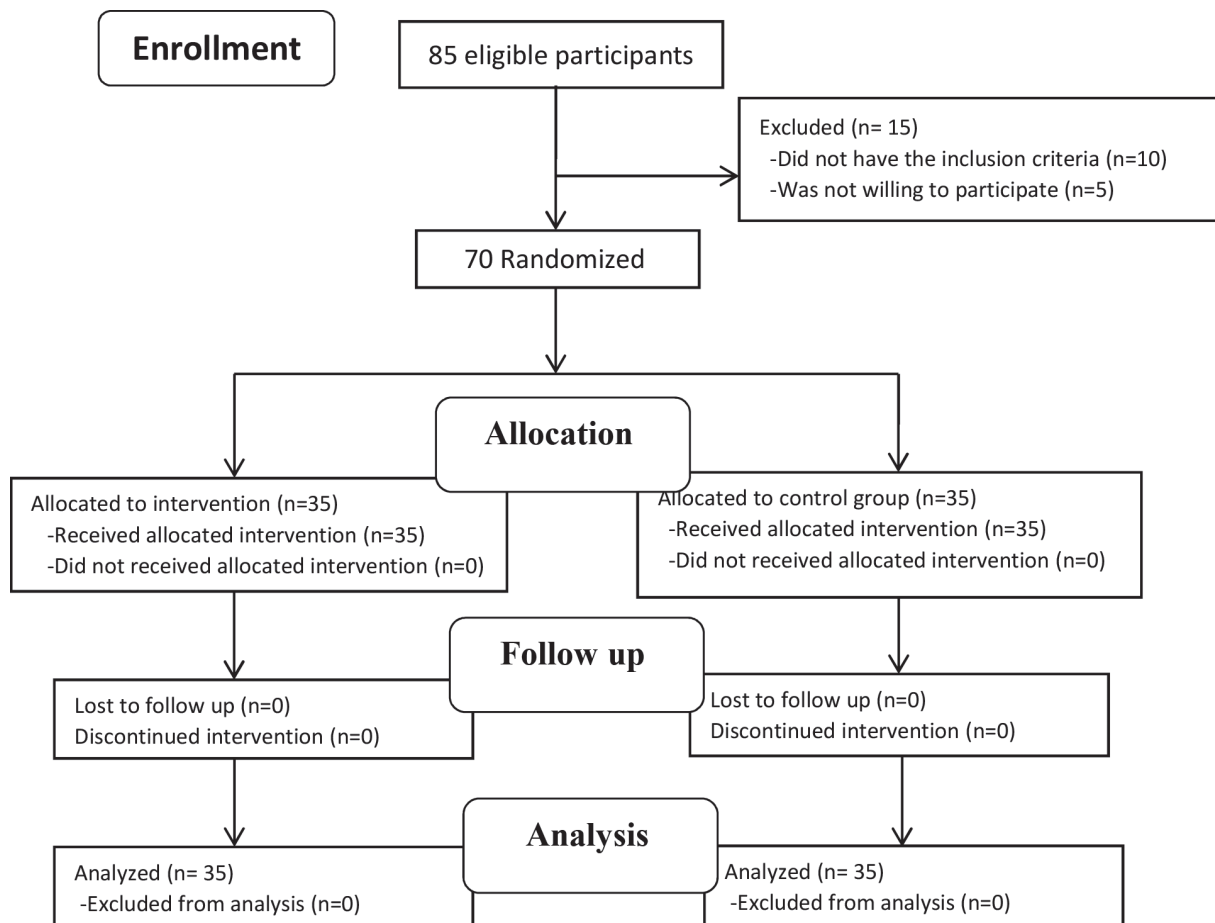


Figure 1: The flow diagram of the caregivers of the elderly with dementia through each stage of the study

and educational imagery were taught for the intervention group (Table 1). Teaching special topics was done by senior lecturers, neurologists and researchers in each topic. The stabilizing of information provided by the researcher to the family caregivers of the intervention group during the 4-week training sessions, so that the researcher provided caregivers of the intervention group with the contact information and it was requested to make contact by telephone, SMS, video calling with virtual messaging programs or email three hours a day (from 18:00 to 21:00) and receive advisory services. If the patient required more specialized care, the caregiver was advised to refer to a specialist.

After the end of the intervention (four weeks of educational intervention), the second stage of data collection was carried out by completing the Spielberger anxiety questionnaire by the caregivers of the intervention and control groups. Also, to

assess the durability of the effect of education, three months after the intervention, the third stage of data collection from family caregivers in both groups was done.

Demographic and Spielberger questionnaires were used in this study. The demographic questionnaire was self-made and included age, sex, marital status, etc. The State-Trait Anxiety Inventory (STAI) is a commonly used measure of trait and state anxiety. It can be used in clinical settings to diagnose anxiety and also often used in research as an indicator of caregiver distress. This questionnaire has 40 questions, 20 of which evaluate situational anxiety and 20 of them assess personality anxiety. The total score of each of the two scales of situational and personality anxiety is from 20 to 80. The scores of anxiety in Spielberger questionnaire are shown in Table 2. This tool is a valid questionnaire and has been used extensively in domestic and international researches.¹⁸ Spielberger et al.

Table 1: The contents of informational interventions in each session

Session No.	Educational caring type	Content of information	Teaching method technique
1	Patient care (elder with dementia)	General definition of the types of dementia	Discussion
2		Dementia treatments and drugs	Lecture, discussion, questioning strategies
3		Standard cares for the elderly with dementia	Lecture, discussion, questioning strategies
4		Physical and cognitive rehabilitation	Lecture, discussion, questioning strategies
5	Self-care (Caregiver)	Promoting adaptation and coping with the crisis situations and caring problems	Lecture, discussion, questioning strategies
6		Stress management methods	Role play, discussion
7		Promoting quality of life methods in family caregivers of elderly people with dementia	Discussion, questioning strategies, problem solving

(1970) state that for State Anxiety Scale, the stability coefficients range from 0.16-0.62, which are relatively low figures because a valid measure of State Anxiety Scale should reflect the influence of unique situational factors that may exist at the time of testing. The psychometric properties of the inventory in Iranian population were confirmed by exploratory and confirmatory factor analysis. The correlation between the inventory and clinical specialists' assessment also showed appropriate validity.¹⁹ The reliability of the Iranian form of the inventory was Cronbach's alpha 0.88.^{20, 21}

Data collected from this study were analyzed by SPSS version 21 using descriptive (mean and frequency) and analytical statistics (ANOVA test with repeated data and independent t-test).

RESULTS

The normal distribution of variables was verified using Kolmogorov-Smirnov test. Findings showed that most of the family caregivers of the elderly in the intervention group 24 (68.6%) and control group 25 (71.4%) were females. Most of the caregivers 61 (87.1%) were children of the elderly persons in both groups. 24 (68.6%) caregivers in the intervention group and 29 (82.9%) in the control group were married. 29 (82.9%) of the subjects in the control group and 33 (93.4%) in the intervention group had a high school diploma and higher. 20 (57.1%) of the

Table 2: The scores of anxiety in Spielberger questionnaire

Normal	<19
Mild	20-31
Moderate to downward	32-42
Moderate to upward	43-53
Relatively sever	54-64
Sever	65-75
Very severe	>76

family caregivers of the elderly with dementia in the intervention group and 17 (48.6%) of the control group were employed and had an income.

There was no significant difference between the two groups in demographic characteristics of the caregivers of the elderly people ($P>0.05$). The mean age of the family caregivers was 46.8 ± 9.9 years in the two groups. The mean age of elderly people was 73.6 ± 6.96 years in the two groups and the mean duration of dementia in the elderly with dementia was 2.67 ± 1.09 years. There was no significant difference in the age of caregivers ($P=0.446$), age of the elderly ($P=0.425$) and duration of dementia ($P=0.069$) in comparison of the two intervention and control groups. As a result, the two groups were identical regarding the age of the caregiver, age of the elderly and duration of dementia in them.

The mean anxiety scores of the family caregivers in the intervention group before the intervention, one and three months after the intervention, were 74.11 ± 4.73 , 70.51 ± 3.78 and 70.31 ± 3.43 , respectively. Also, the mean anxiety score of the caregivers of the control

Table 3: Comparison of the mean of anxiety score of the family caregivers of the elderly with dementia in the two groups of intervention and control before the intervention, one and three months after the intervention

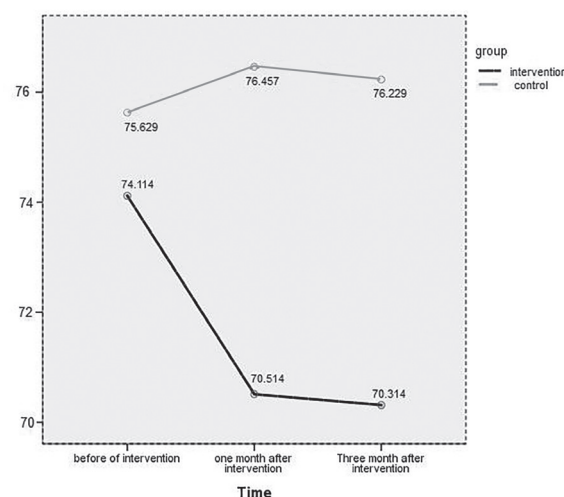
Anxiety	Before the intervention Mean±SD	One month after Intervention Mean±SD	Three months after Intervention Mean±SD	*P value (within)
Intervention	74.11±4.73	70.51±3.78	70.31±3.43	<0.001
Control	75.63±3.08	76.45±3.45	76.22±5.08	0.14
P value (between)	0.12	<0.001	<0.001	
*ANOVA repeated measurements				

group before and one and three months after the intervention were 75.63 ± 3.08 , 76.45 ± 3.45 and 76.22 ± 5.08 , respectively; there was no significant difference in anxiety before the intervention in the two groups (Table 3).

ANOVA test with repeated measurements showed that there was a significant difference between the two groups in terms of the anxiety mean score one month and three months after the intervention ($P < 0.05$) (Table 3). This finding suggested that the reduction in anxiety in the intervention group was significant compared to the control group (Figure 2).

DISCUSSION

The present study showed that the informational support of the caregivers of the elderly with dementia could reduce anxiety in caregivers. Education and care could reduce the anxiety of family caregivers of the elderly patients with Alzheimer's disease.²² Family education programs could improve the psychological situation of family caregivers of the elderly with dementia and reduce the caregivers' anxiety.²³ Caregivers of the elderly patients with Alzheimer's disease were not in a good general health and the educational counseling program could significantly improve the general health of the caregivers.²⁴ The family education program reduced anxiety in family caregivers of mental disorder patients and could potentially improve the quality of life of the caregivers and patients.²⁵ Also, this increased their satisfaction and awareness about Alzheimer's disease, which reduced stress. Caregivers who had practiced different types of adaptive mechanisms during the educational intervention reported more

**Figure 2:** The mean changes in anxiety scores of family caregivers of the elderly patients with dementia before, one and three months after the intervention, by two groups of intervention and control

favorable physical health and satisfaction.²⁶

Most caregivers were female and married. In Iran and in other countries, caring for chronic patients, especially the elderly, is done by women who are middle-aged and married.²⁷ In a study by the National Caregivers Union and the American Association of Retired Persons was found that more than half of the caregivers of patients with chronic disorders were middle-aged women with a mean age of 46 years.^{20, 21} These findings were in agreement with the present study. Previous studies have also shown that the less caregiver used adaptation methods to get adapted to the problems caused by illness, the more vulnerable they were to physical and psychological complications (anxiety and depression).²⁸

The results of the present study were in the same line with those of previous researches on the effectiveness of informational support

on caregivers; however, what distinguished this study from other studies and made it unique was the existence of a non-intrusive in-person informational support program that was compiled and structured that caregivers, using easy and varied communication methods, shared care challenges at home with the researcher and surveyed the problems with the elderly care with the help of the researcher and at home to solve the problem. The use of various educational materials, such as using books and new and varied educational methods, in part of each session, was another advantage of this study. In this study and in most of the previous studies, the duration of the follow-up was short. Therefore, regarding the chronicity of dementia, family caregivers need longer support for the care of the elderly with dementia. Thus, considering the limitations and results of the study, it is suggested that the effect of more comprehensive psychological-spiritual programs such as emotional coping (dancing, deep breathing, yoga, relaxation, etc.) on the anxiety of caregivers should be investigated. It is also recommended to conduct more studies with larger sample size and longer follow-up time.

Due to the homogeneity of demographic data in both groups, these factors cannot be identified as confounding in this study. In addition, due to the generalization limitations of the results of the available sampling method, education and informational support is provided by the relevant organizations on a wider range of caregivers.

Nurses working in health centers, as well as sociologist nurses, other elderly people and home care centers need to receive the knowledge and skills necessary about threatening situations of the health of the elderly to maintain and promote the health and improve performance of caregivers and their elderly relative in all levels, especially family and community.

CONCLUSION

Anxiety among caregivers of the elderly with

dementia is in high level, so according to the findings of this study, educational intervention can affect the family caregivers of dementia elderly and reduce their anxiety. Furthermore, many caregivers require more trainings and support as well as physical and mental health care. Therefore, provision of educational services for family caregivers of the elderly people with dementia is essential and should be specifically addressed in medical systems.

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Conflict of Interest: None declared.

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Clinical Trial Protocol

Iranian Registry of Clinical Trials

08 Dec 2022

The effects of informational support on anxiety and quality of life in family caregivers of elders with dementia

Protocol summary

Summary

The empowerment and support of family caregivers is important; so the aim of this study is to investigate the effect of information support on anxiety and quality of life of family caregivers with dementia elderly. In this quasi-experimental and interventional two-grouped and three-stage study, 70 family caregivers with dementia elderly ask to participate through a call in neurologic and psychologic clinics affiliated to Shiraz University of Medical Sciences. Caregivers should be above 18 years old, experience in patient care for at least three consecutive months, and access to the mobile and social networks. They should also get average score of anxiety caregivers obtained from Spielberger questionnaire more than 31; and the score of quality of life obtained from Thomas quality of life questionnaire less than 80. Referring the elderly to long-term care centers, the occurrence of a crisis events or severe stress such as the death of loved ones and divorce for the caregiver during the study; and the caregiver's attendance in training sessions on the care of the elderly with dementia over the past 6 months leads to caregiver's exclusion from the study. The caregivers divide into two experimental (35 people) and control (35 people) groups by random block allocation. The experimental group receive information support in two distinct parts including training (elderly with dementia care and self-care in seven 90-minute training sessions) and information-counseling support (by phone, SMS, video call with virtual messenger programs or email for 4 weeks, 3 hours a day). Blinding is not done in this study. The control group receive no information support during the study. In this study, family caregiver's anxiety and quality of life measure, compare and evaluate in three stages: pre-study, immediately after the last counseling session and three months later.

General information

Acronym

IRCT registration information

IRCT registration number: **IRCT2017080915426N4**

Registration date: **2017-10-03, 1396/07/11**

Registration timing: **registered_while_recruiting**

Last update:

Update count: **0**

Registration date

2017-10-03, 1396/07/11

Registrant information

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Name of organization / entity

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Recruitment status

Recruitment complete

Funding source

Shiraz University of Medical Sciences

Expected recruitment start date

2017-07-23, 1396/05/01

Expected recruitment end date

2017-11-22, 1396/09/01

Actual recruitment start date

empty

Actual recruitment end date

empty

Trial completion date

empty

Scientific title

The effects of informational support on anxiety and quality of life in family caregivers of elders with dementia

Public title

Effects of informational support on elder family caregivers

Purpose

Supportive

Inclusion/Exclusion criteria

• Inclusion criteria: The caregiver's willingness to participate in the research and to complete the self-consent form; taking the main responsibilities towards the patient; having the caregiver's reading and writing skills; being the caregiver's age above than 18 years old; the caregiver's access to phone, mobile, and social networks; experience in patient care for at least three consecutive months; having at least six months to five years dementia disease history; getting average score of anxiety caregivers obtained from Spielberger questionnaire more than 31; and the score of quality of life obtained from Thomas caregiver quality of life questionnaire less than 80. • Exclusion criteria: The caregiver physical and mental illnesses during the study; taking liquors or psychotropic drugs by the caregiver; referring the patient to long-term care centers during the study; the occurrence of a crisis events or severe stress such as the death of loved ones and divorce for the caregiver during the study; the caregiver's attendance in training sessions on the care of the elderly with dementia over the past 6 months; the elderly's suffering from other disabling diseases such as heart failure, asthma or uncontrolled diabetes before or during the study; the caregiver's responsibility of taking care of someone else in the family.

Age

From **18 years** old

Gender

Both

Phase

N/A

Groups that have been masked

No information

Sample size

Target sample size: **70**

Randomization (investigator's opinion)

Randomized

Randomization description

Blinding (investigator's opinion)

Not blinded

Blinding description

Placebo

Not used

Assignment

Parallel

Other design features

Secondary Ids

empty

Ethics committees

1

Ethics committee

Name of ethics committee

Ethics committee of Shiraz University of Medical Sciences

Street address

Zand street

City

Shiraz

Postal code

71348-14336

Approval date

2017-07-19, 1396/04/28

Ethics committee reference number

IR.SUMS.REC.1396.76

Health conditions studied

1

Description of health condition studied

Dementia in Alzheimer disease

ICD-10 code

G30.1

ICD-10 code description

Alzheimer disease with late onset

2

Description of health condition studied

Vascular dementia

ICD-10 code

F01

ICD-10 code description

Vascular dementia

3

Description of health condition studied

Dementia in other diseases

ICD-10 code

F02

ICD-10 code description

Dementia in other diseases classified elsewhere

Primary outcomes

1

Description

Anxiety

Timepoint

Before, exactly and 3 months after intervention

Method of measurement

Spielberger questionnaire

2

Description

Quality Of life
Timepoint
Before, exactly and 3 months after intervention
Method of measurement
Thomas caregivers quality of life questionnaire

Secondary outcomes

empty

Intervention groups

1

Description

The case group receive informative support in two distinct parts, including: in-person training (caring elder with dementia and self-care; in seven 90-minute training sessions) and Informative-counseling support: (by phone, SMS, video call with virtual messenger programs or email for 4 weeks, 3 hours a day). Also a book that approved by university professors give to the caregivers.

Category

Other

2

Description

The control group received no informative support during the study.

Category

Other

Recruitment centers

1

Recruitment center

Name of recruitment center

Emam Reza clinic

Full name of responsible person

Ganj Karimi

Street address

Namazi square

City

Shiraz

2

Recruitment center

Name of recruitment center

Ebn Sina hospital

Full name of responsible person

Shojaee Ali

Street address

Hafez street

City

Shiraz

3

Recruitment center

Name of recruitment center

Professor Moharrery psychological hospital
Full name of responsible person
Homayoonfar Farzad
Street address
Next to Shiraz agriculture college
City
Shiraz

Sponsors / Funding sources

1

Sponsor

Name of organization / entity

Shiraz University of Medical Sciences

Full name of responsible person

Dr Hashemi Seyed Basir

Street address

Zand street

City

Shiraz

Grant name

Grant code / Reference number

Is the source of funding the same sponsor organization/entity?

Yes

Title of funding source

Shiraz University of Medical Sciences

Proportion provided by this source

100

Public or private sector

empty

Domestic or foreign origin

empty

Category of foreign source of funding

empty

Country of origin

Type of organization providing the funding

empty

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Sharing plan

Deidentified Individual Participant Data Set (IPD)

empty

Study Protocol

empty

Statistical Analysis Plan

empty

Informed Consent Form

empty

Clinical Study Report

empty

Analytic Code

empty

Data Dictionary

empty

The Role of Mutuality and Coping in a Nurse-Led Cognitive Behavioral Intervention on Depressive Symptoms Among Dementia Caregivers

Yuqin Pan, PhD, RN; Ruyi Chen, MS, RN; and DongLiang Yang, MS

ABSTRACT

The current study explored whether mutuality and coping predicted and/or mediated the effect of a nurse-led cognitive behavioral intervention (NLCBI) on depressive symptoms of caregivers of persons with dementia. The intervention group ($n = 56$) received five monthly in-home nurse-led cognitive behavioral sessions and consultation calls after each session. The control group ($n = 56$) received five monthly short general conversations with the nurse interventionist. Questionnaires on study variables and demographics were collected at baseline, end of intervention, and 2-month follow up. Improved mutuality ($\beta = -0.75, p = 0.049$) and active coping ($\beta = -2.06, p = 0.0001$) and decreased passive coping ($\beta = 1.43, p = 0.001$) were found to predict the reduction of depressive symptoms among caregivers in the NLCBI. However, none of these variables mediated the interventional effect. Regular mental health nursing interventions are suggested to focus on enhancing mutuality and active coping and decreasing passive coping to maintain caregivers' mental health.

Targets: Caregivers of persons with dementia.

Intervention Description: Nurse-led cognitive behavioral sessions and subsequent consultation calls.

Mechanisms of Action: Impacted caregivers' reappraisals, thus improving their active coping skills and mutuality and decreasing their passive coping, which directly reduced their depressive symptoms.

Outcomes: Mutuality, active coping, and passive coping played a predicting, but not mediating, role in the effect of the NLCBI.

[Res Gerontol Nurs. 2019; 12(1):44-55.]

Family caregivers of persons with dementia (PWD) often experience depressive symptoms, as dementia care is burdensome and of long duration. Caregiving situations are relatively confined as caregivers need to closely attend their PWD and may become homebound, so dyadic mutuality, a

positive quality of the caregiver-care recipient relationship, is important to the mental health of family caregivers (Park & Schumacher, 2014). However, inter- and/or intrapersonal conflicts may arise amidst the caregiving process, damaging mutuality. In China, although caregiving for PWD is

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primarily a family responsibility according to cultural tradition and legislation, the familial network has diminished with urbanization and the previous family planning policy, and caregiver support systems are still underdeveloped. Thus, family caregivers need effective coping strategies to maintain their mental health.

THEORETICAL FRAMEWORK

Lazarus and Folkman's (1984) Stress and Coping Theory was used to guide the design of the current study. Stress and Coping Theory emphasizes the reciprocal relationship between an individual and his/her surroundings. Stressors are seen as deriving from an imbalance between perceived demands and resources, and *coping* refers to an individual's cognitive and behavioral efforts in managing demands that are appraised as taxing or exceeding an individual's resources (Lazarus & Folkman, 1984). In the process of coping, primary and secondary appraisals converge to determine whether the person–environment transaction is significant for well-being, whereas reappraisals occur based on new information. Lazarus and Folkman (1984) posit that cognitive appraisals in person–environment transactions impact an individual's coping behaviors and health outcomes.

Psychosocial programs such as cognitive behavioral interventions (CBIs) may help adjust caregivers' reappraisals on dementia, role demands, and dyadic relationships within specific social contexts; caregivers' interactions with these situations could be adapted using more positive emotions and active coping skills. Therefore, positive reappraisals within transactions may improve caregivers' coping strategies and dyadic relationships, and consequently promote caregivers' mental health (Zauszniewski, Lekhak, & Musil, 2018).

Mutuality and Coping in the Interventional Effect on Caregiver Depression

The Role of Mutuality. Mutuality, a positive quality of dyadic relationships, was viewed as a support resource for caregivers' mental health (Pan, Jones, & Winslow, 2016). Mutuality has four dimensions: affection, reciprocity, sharing of pleasant activities, and sharing of values (Archbold, Stewart, Greenlick, & Harvath, 1990). Extensive literature indicates that higher mutuality is associated with less burden or role strain, fewer depressive symptoms, and moderation or mediation between stress/caregiving demands and mental health among family caregivers (Crist, Pasvogel, Szalacha, & Finley, 2017; Park & Schumacher, 2014; Yang, Liu, & Shyu, 2014). However, few

studies have been conducted using supportive strategies such as CBIs to facilitate caregivers' mutuality, and some studies with relevant interventions revealed inconsistent effects on mutuality or depression.

Kunik et al. (2017) reported that mutuality, but not depression, improved among dementia caregivers with strategies of enhancing communication and making daily activities pleasant. However, Li, Powers, et al. (2012) did not find significant group differences in either mutuality or depression, although caregivers with an empowerment educational program had less role strain during follow up. Gonzalez, Polansky, Lippa, Gitlin, and Zauszniewski (2014) further reported little effect on these variables after a resourcefulness training program among caregivers of PWD. These studies included both mutuality and depression as dependent variables; thus, the role of mutuality in interventions on caregiver depression is unknown (Rausch, Caljouw, & Es, 2017).

The Role of Coping. Coping is a process in which an individual mobilizes his/her resources in response to demands. Coping strategies vary among individuals and situations but are often categorized as active and passive coping, or problem-focused and emotion-focused coping (Lazarus & Folkman, 1984; Xie, 1998). *Active coping* includes positive emotions and problem-solving strategies, such as finding solutions and seeking help, whereas *passive coping*, similar to dysfunctional coping, includes negative emotions such as avoidance and denial, and behaviors such as substance use and daydreaming (Li, Cooper, Bradley, Shulman, & Livingston, 2012; Roche, MacCann, & Croot, 2016). Evidence showed that more active coping reduced depressive symptoms among caregivers, whereas more passive/dysfunctional coping increased depressive symptoms (Chan & Chui, 2011; Lu, Liu, Wang, & Lou, 2017; Roche, Croot, MacCann, Cramer, & Diehl-Schmid, 2015).

Several studies have explored the effect of cognitive-behavioral therapy (CBT) on caregivers' coping strategies, but results varied. Studies concluded three different results: (a) active coping strategies improved (Fialho, Koenig, dos Santos, Barbosa, & Caramelli, 2012; Mahmoodi, Khani, Banab, & Bagheri, 2016); (b) positive and dysfunctional coping strategies increased (Li, Cooper, Austin, & Livingston, 2013); and (c) active coping strategies increased, whereas passive coping strategies selectively decreased (Losada, Marquez-Gonzalez, & Romero-Moreno, 2011). These inconsistencies raised the question of whether coping strategies play any role in the effect of CBT on caregivers' depressive symptoms.

In addition, findings from several relevant studies were conflicted on how CBT relieved caregiver depression. Gallagher-Thompson, Gray, Dupart, Jimenez, and Thompson's (2008) report showed that CBT worked via skill utilization and helpfulness among non-Hispanic White and Hispanic-Latino dementia caregivers. However, Li et al. (2014) found that the effect of a coping skill intervention was mediated by increased emotion-focused coping among caregivers with higher baseline depression, with neither problem-focused nor dysfunctional coping playing a role. A similar study by Losada et al. (2011) indicated that this effect occurred by modifying caregivers' dysfunctional thoughts. Therefore, as caregivers may respond to interventions differently, more research needs to be conducted to elucidate whether coping is a mediator in successful CBT interventions.

PURPOSE

Little evidence exists regarding the role of mutuality and coping in the effect of nurse-led psychosocial programs on caregivers' mental health. The purpose of the current study was to explore whether mutuality and coping either predict or mediate the effects of a nurse-led CBI (NLCBI) on depressive symptoms of family caregivers of PWD. Nursing strategies and social policies can be developed to maintain caregivers' mental health in countries with similar care burdens and socioeconomic contexts such as those in China.

METHOD

Study Design and Sample

This study was a randomized controlled trial approved by Jin Hua Science and Technology Bureau, including review on human subjects. Family caregivers were recruited via resident and staff referrals from community health centers and neurological units of a tertiary hospital of a city in southeast China. Caregivers were recruited if they: (a) were a primary caregiver; (b) were age 20 or older; (c) were providing care to a family member diagnosed with Alzheimer's disease or dementia (including vascular dementia), or with a Mini-Mental State Examination (MMSE) score ≤ 17 ; (d) were living in communities of the city or surrounding areas; (e) had a depression score on the Center for Epidemiologic Studies Depression Scale (CES-D) of ≥ 10 and < 20 , for better response to the intervention and elimination of medical treatments or therapies; and (f) were able to speak or understand Chinese (Putonghua). Respondents were excluded, as interviewed and screened with medical history and records by the re-

searcher, if they had suicidal tendencies, a history of psychotic disorders, signs of intellectual deficits, and/or were undergoing psychosocial therapies and medication treatments.

Sample Size

G*power achieved a sample size of 98 with an F test, three repeated measurements for two independent groups, effect size = 0.25, power = 0.85, and expected $r = 0.5$ between measurements. Effect size was decided based on a systematic review and meta-analysis that showed a small-to-medium effect of CBT on caregiver depression (Liu, 2017). Whereas most psychosocial studies had an attrition rate of 5% to 30% (Cheng et al., 2017; Losada et al., 2015), the current study used a 15% attrition rate, according to Waldorff et al.'s (2012) 11% to 20% attrition rate at 6- to 12-month follow up. Thus, the sample size was finalized to 112 individuals.

Procedure

Entry permissions of the project to research sites were obtained from authorities of the community health centers and hospital. Flyers and word of mouth were used to explain the project and its recruitment criteria. The researcher (Y.P.) contacted potential respondents after referrals to confirm eligibility, obtain verbal consent, and help arrange an appointment for the baseline survey. Five trained research assistants conducted face-to-face surveys anonymously at Time 1 (*baseline*), Time 2 (*end of the 5-month intervention*), and Time 3 (*2-month follow up*) at caregivers' homes or in a hospital unit, as convenient. Research assistants were trained with survey procedures and interview skills, and were provided answers for possible questions. They were qualified to conduct the survey after successfully demonstrating the procedures as supervised by the researcher. Caregivers were asked to fill out questionnaires, and research assistants helped caregivers read the questionnaire if needed. Patients' MMSE was assessed by research assistants.

Questionnaires were collected upon completion. After the initial assessment, caregivers were randomly allocated by a data processor into either the intervention or control group via a computerized random number, with 56 caregivers in each group. The researcher was initially blinded to the study allocation but supervised the intervention; data collectors were blinded to the treatment assignments; and caregivers received a sealed letter with an allocated number and were informed of their treatment when the intervention began. Caregivers were assigned to one of

the five trained nurse interventionists by geographic location. Masking was not possible for the nurse interventionists. Collected data were locked in the researcher's office drawer, and the data file was encrypted and stored on the researcher's personal computer.

Cognitive Behavioral Intervention Group and Control Group

Intervention Group. This NLCBI comprised five monthly 60-minute individual sessions conducted at caregivers' homes at the beginning of each month by appointment, with a 20- to 30-minute consultation call in the middle of the month. Duration of the intervention and follow up was determined based on the time of an interventional effect and retention of respondents. Five RNs were trained on the intervention protocol and psychosocial communication techniques. Although nurses were instructed to follow the protocol, minimal flexibility was allowed to adapt to individual caregivers. A discussion group in an E-chat room using WeChat software was set up for communication among nurse interventionists to ensure consistency in the intervention. At the initial session, caregivers agreed to make up a session within a few days after a missed appointment.

Content of the intervention sessions was constructed and tailored to Chinese caregivers based on the cognitive behavioral strategies introduced by Aboulafia-Brakha, Suchecki, Gouveia-Paulino, Nitrini, and Ptak (2014); Cheng et al. (2017); and Schinköthe and Wilz (2014). Session 1 assessed caregivers' situations and perceptions on caregiving-related issues. Session 2 focused on clarifying caregivers' stressors, appraisals, and coping strategies. Session 3 primarily concentrated on cognitive training by teaching caregivers positive reappraisal strategies, particularly benefit-finding (i.e., viewing challenging situations in positive ways). In Session 4, caregivers were trained in active coping behaviors such as seeking help and nurturing healthy relationships, as well as in avoiding passive coping behaviors. Session 5 provided caregivers with relaxation and self-maintenance skills. Sessions began by building rapport with the caregiver and ended by summarizing the session and assigning homework. Questions used in the face-to-face intervention sessions are listed in **Table 1**.

As mutually agreed, the 20- to 30-minute consultation call was arranged to assess caregivers' recent condition, ask for feedback, and reinforce strategies taught in the sessions. Interventionists were prepared to answer caregivers' questions. Although subsequent sessions still abided by the

protocol, caregivers' feedback was sought to help further clarify details of the intervention. No retention incentives were used, as attrition could partly reflect the effectiveness of the intervention.

Control Group. Caregivers in the control group were provided with five monthly, short (5 to 10 minutes), general conversations with a nurse interventionist at the beginning of each month either at caregivers' homes, at the hospital, or via telephone as convenient. No in-depth instructions were given; conversations were casual chats on caregivers' daily lives and health.

Measures

Depression. The 10-item CES-D was used to assess frequency of depressive symptoms among caregivers. One example of an item was "I felt depressed." Items were rated on a 4-point response format from 0 (*rarely or none of the time*) to 3 (*most of the time or all of the time*). The scale score is the total of the item scores, with ≥ 10 indicating the presence of clinical depression. The Chinese version of the CES-D had a Cronbach's alpha of 0.79 among Chinese caregivers (Pan et al., 2016).

Mutuality. The 15-item Mutuality Scale was used to measure caregiver mutuality. One example of an item was "To what extent do you enjoy the time the two of you spend together?" Items were rated on a 5-point Likert scale ranging from 0 (*not at all*) to 4 (*a great deal*). The scale score is the mean of the item scores; higher scores indicate higher levels of mutuality. The Chinese version of the scale had a Cronbach's alpha of 0.91 among caregivers in China (Pan et al., 2016).

The Simplified Coping Scale. The 20-item Simplified Coping Scale (Xie, 1998) was developed to measure coping strategies of Chinese adults, with a Cronbach's alpha of 0.90 for the full scale, 0.89 for the 12-item subscale of active coping, and 0.78 for the eight-item subscale of passive coping. Examples of items included "Seeking suggestions from friends and relatives" and "Accepting the reality as there are no other choices." Items were rated on a 4-point scale ranging from 0 (*never used*) to 3 (*often used*). The subscale scores are the respective averages of their items. A higher score in active coping indicates better coping, whereas a higher score in passive coping indicates poorer coping.

Care Recipients' Activities of Daily Living. The 14-item Activities of Daily Living (ADL) scale measures the basic and more complicated activities of patients' self-maintenance. One example of an item was "Are you having difficulty eating?" Items were rated on a 4-point scale from 1 (*performed without difficulty*) to 4 (*unable to perform*).

TABLE 1

Key Questions Leading the Face-to-Face Sessions of the Cognitive Behavioral Intervention

Session Number and Objectives	Questions Asked
Session 1 Understand caregivers' situations and perceptions	1. Could you please introduce yourself and describe your caregiving background (socioeconomic status, caregiving history, family network, caregiving demand, and support)? 2. How do you perceive dementia, the person with dementia, your role as a dementia caregiver, your family or dyadic relationship, your available support, and your utilization of this support? 3. Have you felt or experienced the social stigma of dementia?
Session 2 Clarify caregivers' stressors, appraisals, and coping strategies	1. Could you please list a number of difficult caregiving scenarios, your previous management approaches, and consequences of these approaches? 2. Could you please reflect on these scenarios above and identify the stressors, your appraisals, and coping strategies? 3. Are you aware of the positive appraisals and active coping strategies that may have reduced your stress and promoted your problem-solving skills as well as negative appraisals and passive coping strategies that may have increased your stress?
Session 3 Elicit positive appraisals and avoid negative appraisals	1. Could you please recall a couple of the most pleasant or most unhappy caregiving scenarios, and discuss your positive or negative appraisals during the process? For example, do you usually view dementia care and the dyadic relationship as negative or positive? 2. Could you please analyze those pleasant caregiving scenarios and identify the use of the strategy of benefit-finding? For example, have you experienced benefits or gains during the process? 3. Could you please try to apply the strategy of benefit-finding into the unhappy caregiving scenarios, avoid negative emotions, and imagine if the consequences would have been better?
Session 4 Train caregivers to use active coping behaviors and avoid passive coping behaviors	1. Could you please explore the most pleasant or most unhappy caregiving scenarios further, and discuss your active or passive coping behaviors during the process? 2. Could you please try to apply active coping behaviors to the most difficult caregiving scenarios, such as building support, seeking help, and nurturing a good dyadic relationship or a friendly caregiving environment, and imagine if the outcomes would have been better? 3. Would you like to reinforce these active coping behaviors in other challenging caregiving scenarios, and also avoid the use of passive coping behaviors such as substance use and avoidance?
Session 5 Train caregivers on skills for relaxation and self-maintenance	1. What relaxation skills do you use? Please list relaxation skills that might be practical to you. 2. Could you please spend some time now to practice relaxation skills such as deep breathing, writing in a diary, listening to music, exercising, and throwing soft pillows? 3. Do you know how to manage your time, resume pleasant activities or hobbies, balance nutrients, and get better sleep? 4. What other personal skills could you develop that might be beneficial to your health?

The scale score was the sum of the item scores. The Chinese version of the 14-item ADL scale (ADL-C) was used among Chinese family caregivers, with a Cronbach's alpha of 0.97 (Pan et al., 2016).

Care Recipients' Mini Mental State Examination. The Chinese version of the MMSE was used to assess patients' cognitive function. This tool has been widely tested with satisfactory psychometrics and applied to older adults in

China (Yu, Li, & Huang, 2012). The MMSE comprises 10 cognitive function domains, such as memory, orientation, and calculation, with a maximum of 30 points. Higher scores indicate better cognitive function. The cut-off for diagnosing cognitive impairment is 16 to 19 for individuals with primary education and below; thus, a 17-point cut-off was adopted in the current study (Yang et al., 2016).

Demographic and Caregiving Characteristics. A questionnaire designed by the researchers was used to collect demographic data of caregivers and care recipients.

Data Analysis

SPSS 25.0 was used for data analysis. The missForest method (Stekhoven & Bühlmann, 2012) was adopted to replace missing data, including some missing questions from individuals who completed the study and missing questionnaires from those who dropped out, so that the replaced data could best reflect the nature of the true data. Descriptive data were evaluated with frequencies or measures of central tendency. Baseline comparisons in demographics and studied variables between the intervention group and control group, and between individuals who completed the study and those who dropped out, were analyzed using chi-square test, independent sample *t* test, or Mann-Whitney U test, in accordance with the variable type and whether assumptions were met.

Generalized linear mixed model (GLMM) was used to analyze the effect of the NLCBI on caregivers' depressive symptoms, and the predicting role of mutuality, active coping, and passive coping in this interventional effect. This analysis allows both the fixed and random effects to be explored simultaneously. The repeated measures (Time 1, Time 2, and Time 3) of the studied variables were reconstructed. Group, time, the interaction of group and time, and the studied variables were input to the fixed effect model, considering the covariates of caregivers' gender and patients' ADL and MMSE scores based on their significance with baseline depression scores by a primary screen. The time variable was placed into the random model, and the reconstructed depression score was the dependent variable. β and *p* values were presented.

To explore the potential mediating effect of mutuality, active coping, and passive coping, Baron and Kenny's (1986) procedure was adopted as suggested by Murphy, Cooper, Hollon, and Fairburn (2009), and as applied in similar studies (Li, Cooper, et al., 2014; Losada et al., 2011). A mediator needs to meet the following conditions analyzed through respective regression analysis: (a) variations in the independent variable (intervention group) must account for variations in the mediator (mutuality, active coping, or passive coping); (b) variations in the independent variable must account for variations in the dependent variable (depressive symptoms); and (c) while regressing the dependent variable on both the independent variable and the mediator, the independent variable must reduce or even lose its effect on the dependent variable,

while the mediator must sustain its significance in affecting the dependent variable. The changes of the studied variables were calculated respectively by subtracting Time 1 (T_1) scores from Time 2 (T_2) or Time 3 (T_3) scores. The same covariates as in the GLMM were controlled.

RESULTS

The current study was conducted from May 2016 to March 2017 in a city in southeast China. Family caregivers of PWD ($N = 112$; intervention group $n = 56$, control group $n = 56$) were recruited from the 276 caregivers referred and were evaluated at Time 1. At Time 2, 99 caregivers (intervention group $n = 54$, control group $n = 45$) completed the survey; and at Time 3, 82 caregivers (intervention group $n = 47$, control group $n = 35$) completed the full three evaluations. Among dropouts in the intervention group, two caregivers' PWD died, six caregivers discontinued caregiving, and one caregiver relocated; in the control group, seven caregivers' PWD died, three caregivers discontinued caregiving, eight caregivers relocated, and three caregivers refused the repeated surveys.

Demographics of family caregivers and PWD are listed in **Table 2**. No significant differences were found in baseline characteristics between the intervention group and control group. Means and standard deviations of the studied variables are listed in **Table 3**. Significant group differences were indicated by the independent sample *t* test in depressive symptoms and active coping at Time 2 and Time 3, respectively.

In **Table 4**, the fixed model of GLMM revealed that, after controlling for covariates, the NLCBI was effective in reducing caregivers' depressive symptoms over time (group and time interaction, $\beta = 1.12$, $p = 0.0001$). Group alone was almost significant ($\beta = 1.41$, $p = 0.053$) but time was highly significant with reducing depressive symptoms. Both active coping ($\beta = -2.06$, $p = 0.0001$) and passive coping ($\beta = 1.43$, $p = 0.001$) strongly predicted caregivers' depressive symptoms in this NLCBI, whereas mutuality was a weaker predictor ($\beta = -0.75$, $p = 0.049$).

As shown in **Table 5**, none of the T_2T_1 or T_3T_1 changes in active coping played a mediating role in the effect of the NLCBI on the corresponding changes of caregivers' depressive symptoms after covariates were adjusted. The significance of group (the independent variable) was not affected by adding the change of active coping (the potential mediator) to the models (active coping: T_2T_1 [$p = 0.002$], T_3T_1 [$p = 0.028$]; group: $p < 0.0001$ in both models). Mutuality and passive coping did not meet the first condition as a mediator. Group was not a significant variable in

TABLE 2
Participant Characteristics at Baseline

Variable	n (%)			Statistical Test (p Value)
	Total (N = 112)	Intervention Group (n = 56)	Control Group (n = 56)	
Caregivers				
Age (years) (mean [SD])	62.68 (10.86)	63.27 (11.22)	62.09 (10.56)	$t = 0.57$ (0.57)
Gender				$\chi^2 = 2.44$ (0.12)
Female	70 (62.50)	31 (55.36)	39 (69.64)	
Male	42 (37.50)	25 (44.64)	17 (30.36)	
Relationship to care recipient				$\chi^2 = 0.14$ (0.71)
Spouse	54 (48.21)	28 (50)	26 (46.43)	
Other family member	58 (51.79)	28 (50)	30 (53.57)	
Educational level				$\chi^2 = 0.001$ (0.99)
Middle school and below	100 (89.29)	50 (89.29)	50 (89.29)	
High school and above	12 (10.71)	6 (10.71)	6 (10.71)	
Number of chronic diseases (mean [SD])	1.08 (1.13)	1.31 (1.29)	0.86 (0.90)	$Z = -1.89$ (0.07)
Duration of care (months) (mean [SD])	67.25 (75.58)	71.2 (85.57)	63.27 (64.61)	$Z = -0.52$ (0.60)
Care recipients				
Age (years) (mean [SD])	79.48 (9.54)	79.00 (9.35)	79.96 (9.78)	$t = -0.53$ (0.60)
Gender				$\chi^2 = 3.12$ (0.08)
Female	71 (63.39)	40 (71.43)	31 (55.36)	
Male	41 (36.61)	16 (28.57)	25 (44.64)	
ADL score ^a (mean [SD])	43.91 (10.54)	42.93 (10.70)	44.89 (10.38)	$t = -0.98$ (0.33)
MMSE score ^b (mean [SD])	6.54 (5.87)	6.45 (5.87)	6.64 (5.91)	$t = -0.18$ (0.86)

Note. ADL = activities of daily living; MMSE = Mini Mental Status Examination. Z scores were calculated using a Mann-Whitney U test.

^a Scores range from 14 to 56, with higher scores indicating lower ability to perform ADL.

^b Scores range from 0 to 30, with higher scores indicating better cognitive function.

the respective model of T_2T_1 change in mutuality ($\beta = 0.13$, $p = 0.20$) and passive coping ($\beta = -0.04$, $p = 0.69$). The models of their T_3T_1 changes achieved similar results. Thus, further analyses were not necessary.

As the attrition rate was higher in the control group (37.5%) than in the intervention group (16%), comparison analyses between individuals who completed ($n = 82$) and dropped out of ($n = 30$) the intervention were performed in the key baseline characteristics of the dyads. No significant group differences were found in caregiver genders ($\chi^2 = 0.58$, $p = 0.49$), relationship to care recipient ($\chi^2 = 2.19$, $p = 0.20$), or depressive symptoms ($t = 0.57$, $p = 0.57$); or patients' age ($t = -0.23$, $p = 0.82$), gender ($\chi^2 = 0.01$, $p = 0.99$), ADL score (Mann-Whitney

U, $p = 0.051$), and MMSE scores (Mann-Whitney U, $p = 0.70$). However, younger caregivers (mean age = 58.93 years, $SD = 9.08$ years) were more likely to drop out than older caregivers (mean age = 64.05 years, $SD = 11.18$ years; $t = -2.25$, $p = 0.037$).

DISCUSSION

Consistent with findings from the literature (Hopkinson, Reavell, Lane, & Mallikarjun, 2018; Liu, 2017), this NLCBI was found to be therapeutic for caregiver depression, as the favored changes in caregivers' mutuality and coping played a key role. Because few similar studies have been conducted by nurses in the target population, this finding enriches the field of research and informs authorities that CBIs by community

TABLE 3

Means, Standard Deviations, and *t* Scores of Studied Variables Over Time

Variable	Mean (<i>SD</i>)		
	Baseline	End of Intervention	2-Month Follow Up
Mutuality ^a (<i>t</i> test)	0.01	1.47	1.47
Intervention group	1.15 (0.40)	1.33 (0.59)	1.27 (0.54)
Control group	1.15 (0.51)	1.19 (0.42)	1.13 (0.50)
Active coping ^b (<i>t</i> test)	-0.37	4.63***	2.92**
Intervention group	1.15 (0.40)	1.65 (0.45)	1.45 (0.43)
Control group	1.18 (0.37)	1.28 (0.38)	1.24 (0.32)
Passive coping ^b (<i>t</i> test)	-0.21	-0.65	0.54
Intervention group	1.38 (0.42)	1.24 (0.38)	1.13 (0.39)
Control group	1.39 (0.35)	1.29 (0.32)	1.09 (0.30)
Depressive symptoms ^c (<i>t</i> test)	1.15	-4.00***	-3.35**
Intervention group	13.93 (3.45)	9.40 (3.39)	10.19 (2.78)
Control group	13.21 (3.11)	11.96 (3.40)	12.21 (3.56)

Note. Data were based on $N = 112$, intervention group $n = 56$, and control group $n = 56$. Actual sample size at end of intervention was $N = 99$, intervention group $n = 54$, and control group $n = 45$. Actual sample size at 2-month follow up was $N = 82$, intervention group $n = 47$, and control group $n = 35$. Missing data were replaced using the missForest method.

^a Measured using the 15-item Mutuality Scale. Item scores range from 0 to 4, with higher scores indicating higher levels of mutuality.

^b Measured using the 20-item Simplified Coping Scale, with item scores ranging from 0 to 3. A higher score in active coping (12 items) indicates better coping, whereas a higher score in passive coping (8 items) indicates poorer coping.

^c Measured using the 10-item Center for Epidemiologic Studies Depression Scale. Scores range from 0 to 30, with higher scores indicating more depressive symptoms.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

nurses can maintain caregivers' mental health in countries that lack adequate numbers of primary practitioners.

The Role of the Studied Variables in the Interventional Effect

Mutuality. Mutuality was indicated as a significant, although weak, predictor of caregivers' depressive symptoms in the NLCBI after covariates were adjusted. Cognitive training has been found to enhance caregivers' appraisals on dyadic mutuality (Li, Powers, et al., 2012), and higher mutuality is known to mitigate caregiver depression (Crist et al., 2017; Ostwald et al., 2014). However, there was only a slight increase in caregivers' mutuality in the NLCBI group, which was in line with the finding of Kunik et al. (2017) but contrary to findings of Gonzalez et al. (2014) and Orrell et al. (2017). These differences could be due to the strategic diversity among similar studies: for example, the current NLCBI did not fully focus on mutuality but included it as one aspect of the intervention.

In addition, the small increase in mutuality helped explain its non-mediating role in the effectiveness of the NLCBI, with some contributing factors. PWD in the current study with poor cognitive function may have lost their

ability to interact or reciprocate in depth, which could have compromised caregivers' mutuality (Park & Schumacher, 2014). Moreover, caregivers in the current study were moderately depressed, which may have decreased their mutuality compared to caregivers who were not depressed (Shim, Landerman, & Davis, 2011); this could explain caregivers' sluggish response to the NLCBI. Although this finding adds to knowledge of the topic, future investigations could focus the intervention on mutuality among caregivers of PWD with less cognitive impairment, therefore helping better understand the dynamics of mutuality and its roles.

Active Coping. The current study showed that, after adjusting for covariates, enhancement in active coping strongly predicted fewer depressive symptoms among caregivers with NLCBI, a finding supported by similar studies (Cheng et al., 2017; Li et al., 2014). This NLCBI partly dealt with caregivers' perceptions of caregiving demands and cultural barriers, such as the social stigma of dementia and caregivers' reluctance to seek help within the social context of China. Caregivers' reappraisals on the perspectives above may have grown more positive, and their active coping strategies improved, resulting in reduced caregiver

TABLE 4

The Fixed Effects of Mutuality, Active Coping, and Passive Coping on Caregivers' Depressive Symptoms in a Generalized Linear Mixed Model (N = 112)

Predictor Variables	Depressive Symptoms	
	β	p Value
Intercept	3.18	0.18
Caregiver gender	1.02	0.044*
Patients' Activities of Daily Living score	0.07	0.009**
Patients' Mini-Mental State Examination score	0.13	0.007**
Group	1.41	0.053
Time ^a		
End of intervention	3.70	0.0001***
2-month follow up	4.93	0.0001***
Group $\times$ time	1.12	0.0001***
Mutuality	-0.75	0.049*
Active coping	-2.06	0.0001***
Passive coping	1.43	0.001**

Note. Independent variables of mutuality, active coping, and passive coping, and the dependent variable of depressive symptoms, are the reconstructed variables with their baseline, end of intervention, and 2-month follow up scores combined.

^a Baseline time was used as reference group.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

depression (Lazarus & Folkman, 1984). Thus, regular, tailored mental health nursing interventions such as this NLCBI could be useful in relieving caregivers' mental distress in developing countries.

However, active coping was not found to play a mediating role in the interventional effect in the current study, which contradicted findings of Gallagher-Thompson et al. (2008) but partially aligned with Li et al.'s (2014) report. The current study did not include the mediating variable of skill utilization as in previous studies, and emotional coping was not measured separately from behavioral coping; however, coping strategies included in these mediators could be reflected in the scale of active coping in the current study. Varied research designs including the use of different scales may have contributed to inconsistencies among these studies. Thus, the mediating role of active coping could be clarified by repeating this study.

Passive Coping. The decrease in passive coping predicted fewer depressive symptoms among caregivers receiving the NLCBI, with covariates adjusted. As shown by López-Martínez (2008), caregivers who received psychosocial interventions perceived higher social support, and, as a result, responded with fewer passive coping strategies. Nevertheless, the decrease in passive coping in this study was contrary to the finding from a similar study by Li et al. (2014) but consistent with Losada et al.'s (2015) finding. These different results could be due to caregiver responses from diversified cultures, or the use of mixed scales: passive coping, dysfunctional coping, or emotion-focused coping.

Moreover, changes in passive coping were not likely to play a mediating role in the interventional effect, as the mediating conditions were not met. It is possible that this NLCBI focused more on benefit-finding, which mainly trained caregivers to have positive thoughts and, consequently, to use more active coping strategies, which may have had less of an impact on passive coping. That said, it is worth noting that strategies of both groups (intervention and control) may be equally effective in affecting caregivers' passive coping (as indicated by *t* test). Therefore, the current study contributed first-hand knowledge of how a decrease in passive coping alleviated depression through mental health nursing interventions among Chinese caregivers. However, this mechanism deserves further confirmation in the target population due to the few references available.

LIMITATIONS

The current study explored a rarely investigated topic—the predicting and mediating roles of mutuality and coping in the effectiveness of a NLCBI on dementia caregivers in China. However, there were limitations in this study.

First, although missing data were replaced by the missForest method, the high attrition rate in the control group and the age difference between individuals who completed and dropped out of the study may have compromised the final results. Caregivers in this study who were depressed were inclined to discontinue caregiving; younger caregivers in China may have other competing roles such as child rearing and were prone to relocate; and PWD in poor condition often did not live through the three survey stages. Future studies should select respondents who would be more likely to complete the study and develop incentives to retain caregivers.

Second, a no-treatment comparison group was not included in this study. It is possible that part of the

TABLE 5
Analysis of the Mediation of Active Coping (N = 112)

	Independent Variable	Dependent Variable	β	t Test	p Value
Regression Analysis	Modules for T ₂ T ₁ Changes				
	Intervention group	Change in active coping	0.41	4.58	0.0001***
	Intervention group	Change in depressive symptoms	-0.48	-5.58	0.0001***
	Intervention group	Change in depressive symptoms	-0.36	-4.03	0.0001***
	Change in active coping		-0.28	-3.17	0.002**
	Modules for T ₃ T ₁ Changes				
	Intervention group	Change in active coping	0.27	2.88	0.005**
	Intervention group	Change in depressive symptoms	-0.39	-4.29	0.0001***
	Intervention group	Change in depressive symptoms	-0.33	-3.61	0.0001***
	Change in active coping		-0.21	-2.22	0.028*

Note: T₁ = Time 1 (baseline); T₂ = Time 2 (end of intervention); T₃ = Time 3 (2-month follow up). T₂T₁ = T₂ Score - T₁ Score; T₃T₁ = T₃ Score - T₁ Score. All regression models controlled for caregiver gender, patients' Activities of Daily Living score, and Mini Mental Status Examination score.

*p < 0.05; **p < 0.01; ***p < 0.001.

interventional effect was due solely to the nurses' attention; in addition, relevant variables such as perceived social support were not included, which may have resulted in biased findings. Therefore, it is suggested that future research designs contain a no-treatment group for observation and a confounding variable of perceived social support.

IMPLICATIONS AND DIRECTION FOR FUTURE RESEARCH

Several implications were derived from the current research. This study partially supported the Stress and Coping Theory by Lazarus and Folkman (1984), as active coping was found to act as a predictor, but not a mediator, in the interventional effect. This NLCBI most likely worked on caregivers' reappraisals, which in turn improved caregivers' active coping skills and mutuality, decreased their passive coping, and ultimately reduced their depressive symptoms.

In relation to nursing practice, this NLCBI, which featured both face-to-face sessions and telephone consultations, was beneficial in reducing the mental

stress of depressed, homebound caregivers of PWD. This approach can be further developed, and community nurses should be trained in its use.

As for nursing research, a similar research design could be conducted with a longer observation period while controlling for the attrition rate. Studies aiming to improve caregiver mutuality warrant further exploration, and the mediating role of mutuality or coping needs to be confirmed. Finally, perceived social support can be included as a potential mediator.

China has reached a critical stage in its social policies about dementia care, and it is time to acknowledge family caregivers' contributions. Supporting family caregivers will have a significant effect because caregivers' well-being helps sustain family caregiving to PWD, which in turn greatly reduces the nation's financial health care investments. Thus, mental health nursing services such as NLCBIs should be initiated to meet caregivers' psychological needs. These services may also be combined with other social policies to combat the crisis of dementia care.

CONCLUSION

Improved mutuality and active coping and reduced passive coping played a predicting, but not mediating, role in the NLCBI on relieving depressive symptoms among family caregivers of PWD. This study contributed to preliminary findings on the mechanisms of the intervention. Mental health nursing programs and supportive social policies are suggested to improve caregivers' mutuality and active coping, and to reduce passive coping to maintain caregivers' mental health in countries with similar socioeconomic contexts to China.

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